Nebico Oil Retention Project (3.1)

Coolmay L10S Ladder program
TK8050 x 3, TK8100 x 1

Modifications:

1. Pointer Functions to shorten bulk calculation
2. Revised Core logic (Mode Scheduler)
3. Revised Oil Distribution
4. Clean Segmentation

Prepared By: Sambridda Ranabhat (Architect & Programmer)

09/09/2026
Mechatronics Engineering .pvt .ltd

Data Name : Memory Capacity
Memory Capacity

[Memory Capacity]
16000

[Program Capacity]
16000 Steps

[Comments Capacity]
0 Block 0 Points

[File Register Capacity]
0 Block 0 Points

[Special Function Block Setting]
0 Block

[Positioning]
0 Block

[Built-in CC-Link/LT Setting]
0 Block

Data Name : PLC System
PLC System

[Battery Less Mode]
OFF

[MODEM Initialized]
None

[RUN Terminal Input]
None

[Device]

	Sym.	Dig.	Points	Start	End	Latch Start	End	Latch Setting Range
Supplemental Relay	M	10	7680	0	7679	500	1023	0 - 1023
State	S	10	4096	0	4095	500	999	0 - 999
Timer	T	10	512	0	511			
Counter(16bit)	C	10	200	0	199	100	199	0 - 199
Counter(32bit)	C	10	56	200	255	220	255	200 - 255
Data Register	D	10	8000	0	7999	200	511	0 - 511
Extended Register	R	10	32768	0	32767			

CC-Link Setting

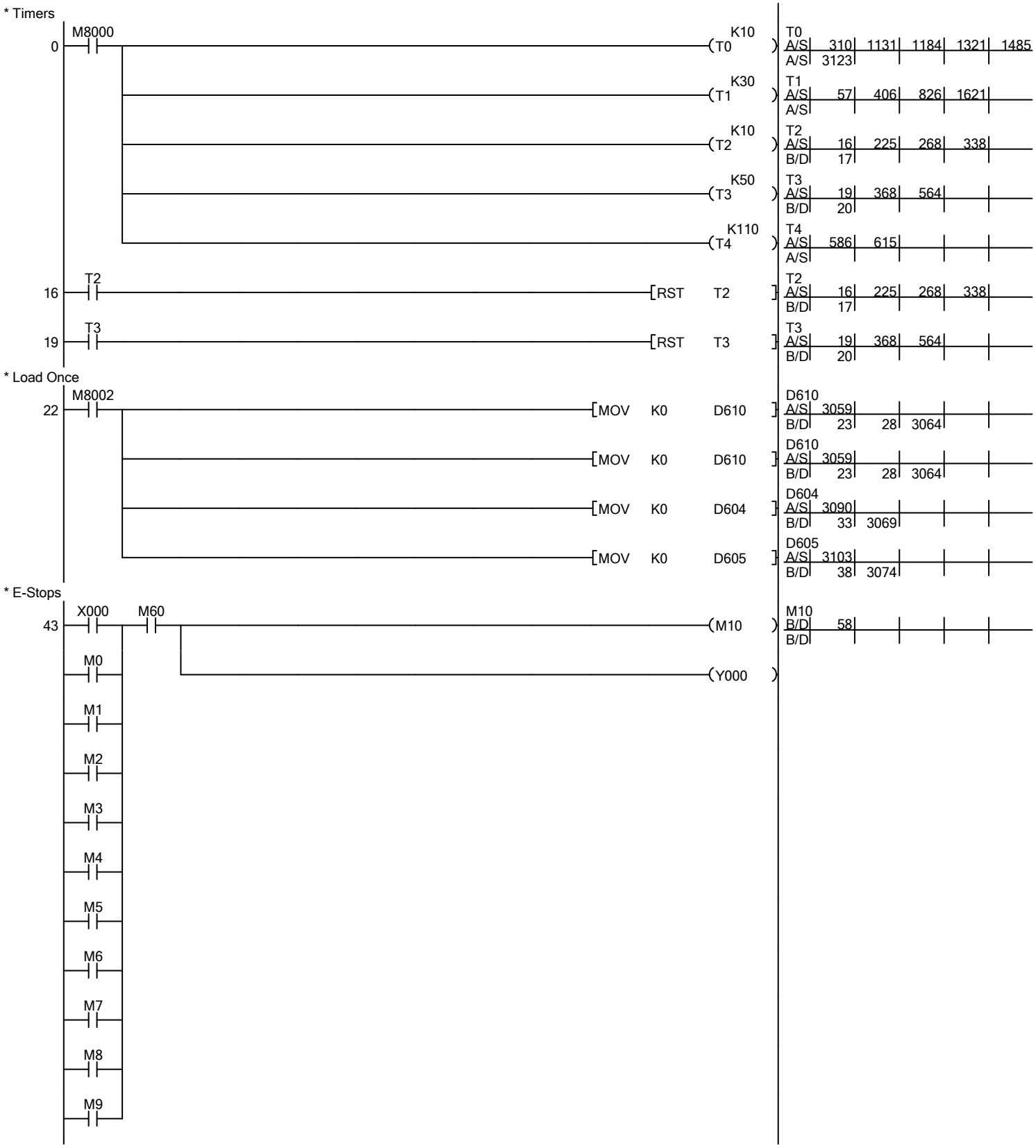
[CC-Link Setting]

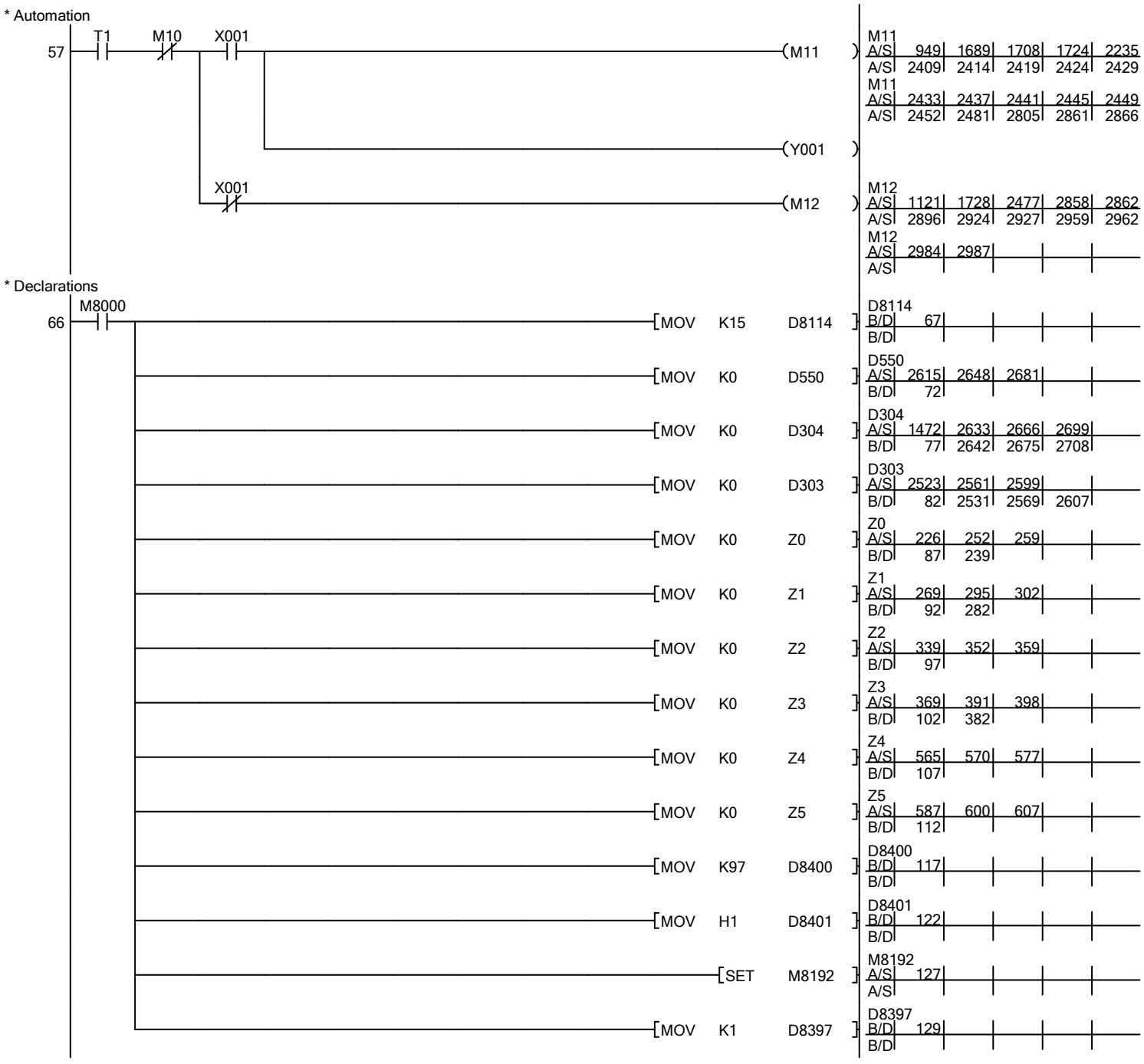
Connection Block Not Set

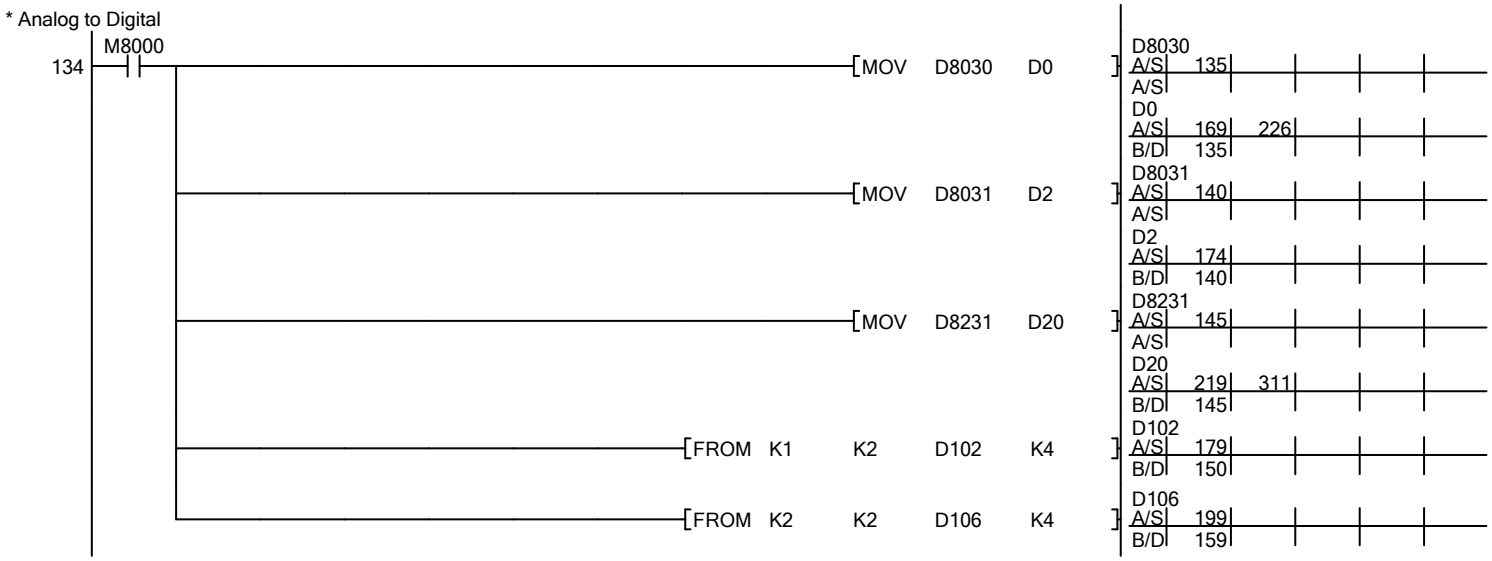
Data Name : CC-Link
CC-Link Setting

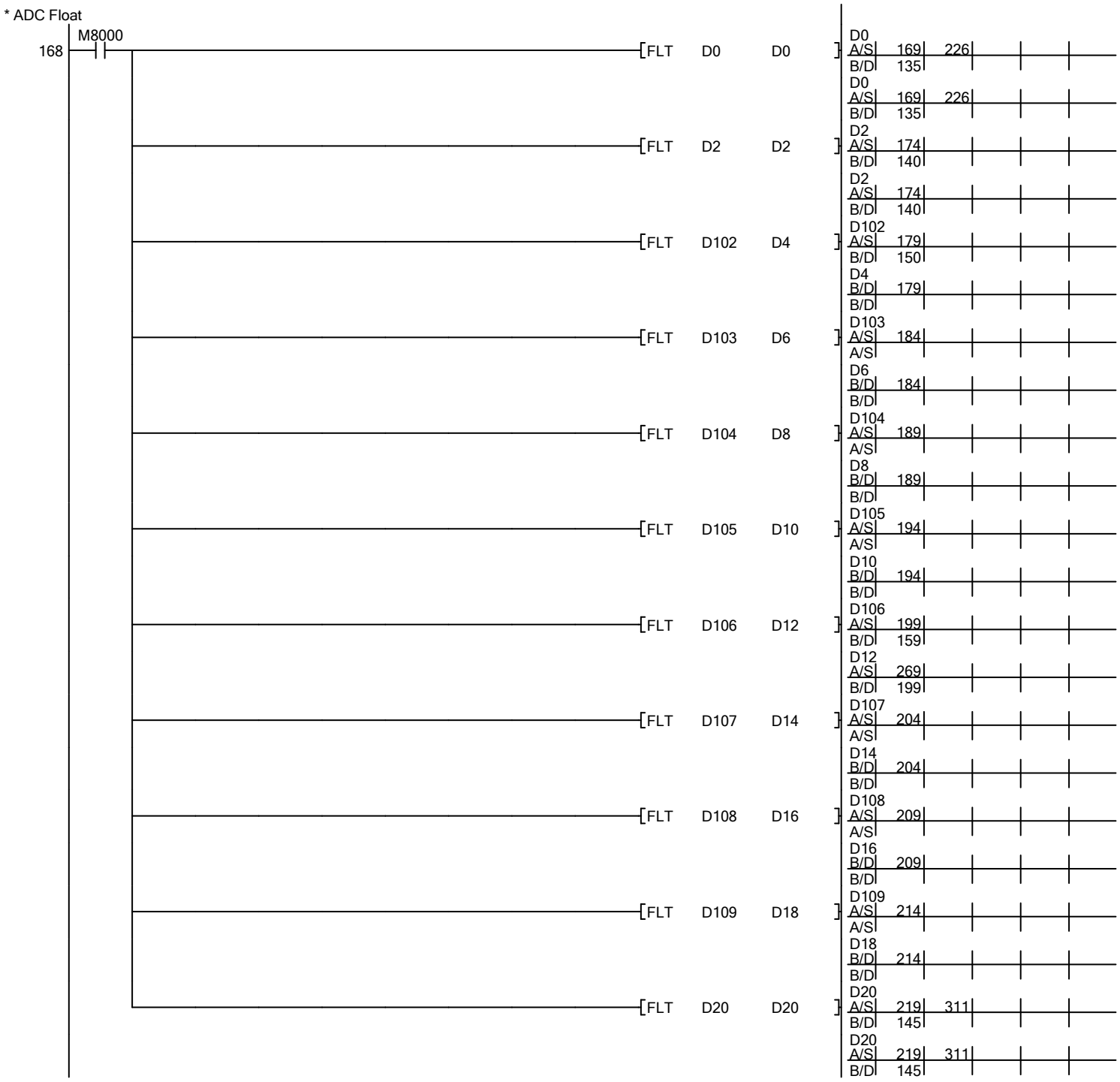
[CC-Link Setting]

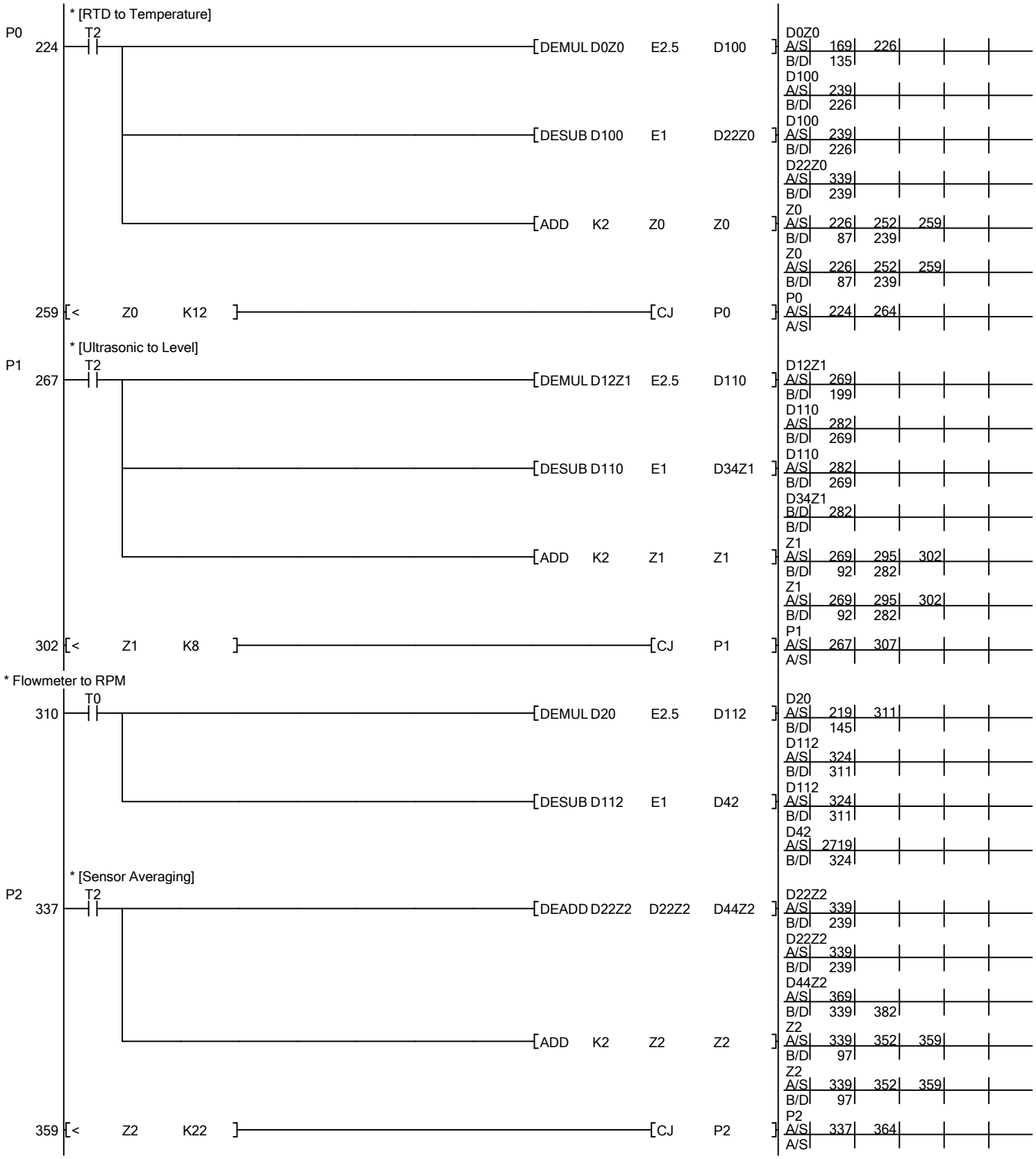
	Setting Contents
Special Function Block No.	-
Type	-
Master Station Data Link Type	Start Parameter by BFM
Mode	-
Total Module Connected	-
Remote Input(RX)	-
Remote Output(RY)	-
Remote Register(RWr)	-
Remote Register(RWw)	-
Ver.2 Remote Input(RX)	-
Ver.2 Remote Output(RY)	-
Ver.2 Remote Register(RWr)	-
Ver.2 Remote Register(RWw)	-
Special Relay(SB)	-
Special Register(SW)	-
Retry Count	-
Automatic Reconnection Station Count	-
Standby Master Station No.	-
PLC Down Select	-
Scan Mode Setting	-
Delay Time Setting	-
Remote Device Station Initial Setting	-
Interrupt Settings	-

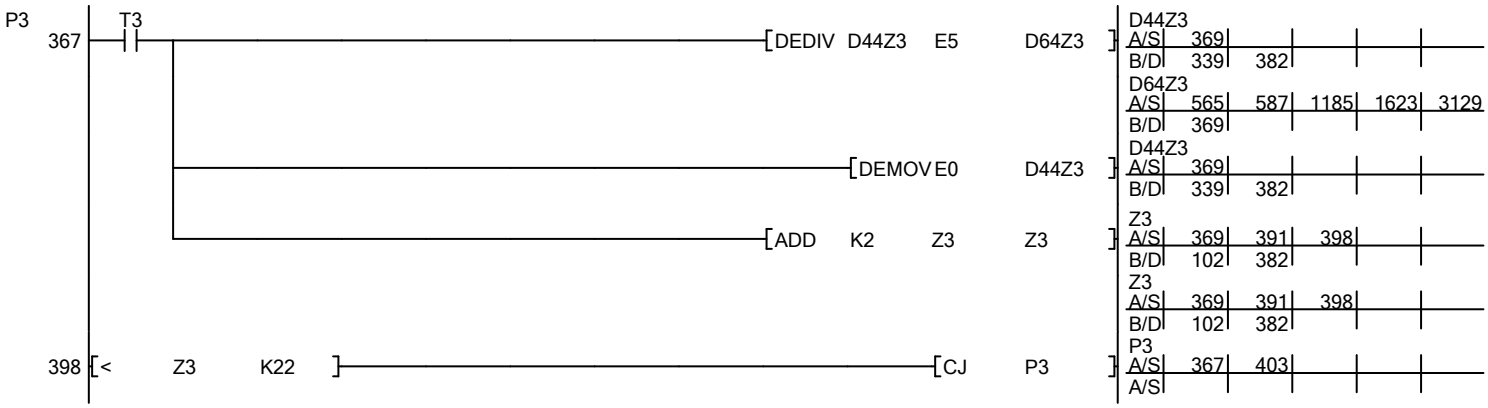

* Timers
0
M8000
K10
T0
T0
A/S 310 1131 1184 1321 1485
A/S 3123
K30
T1
T1
A/S 57 406 826 1621
A/S
K10
T2
T2
A/S 16 225 268 338
B/D 17
K50
T3
T3
A/S 19 368 564
B/D 20
K110
T4
T4
A/S 586 615
A/S
16
T2
RST T2
T2
A/S 16 225 268 338
B/D 17
19
T3
RST T3
T3
A/S 19 368 564
B/D 20
* Load Once
22
M8002
MOV K0 D610
D610
A/S 3059
B/D 23 28 3064
MOV K0 D610
D610
A/S 3059
B/D 23 28 3064
MOV K0 D604
D604
A/S 3090
B/D 33 3069
MOV K0 D605
D605
A/S 3103
B/D 38 3074
* E-Stops
43
X000
M60
M10
M10
B/D 58
B/D
Y000
M0
M1
M2
M3
M4
M5
M6
M7
M8
M9

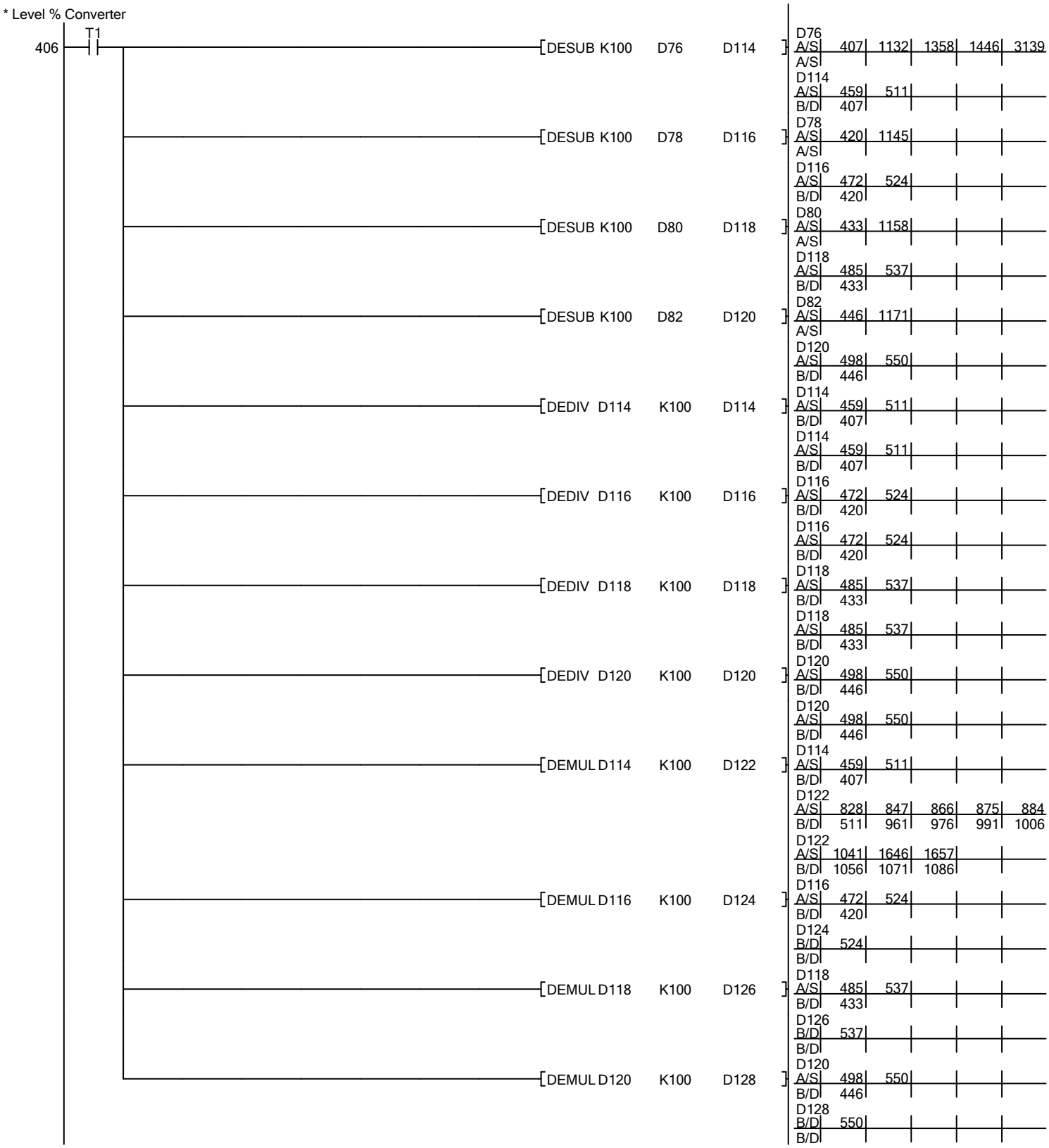

* Automation
57
T1
M10
X001
M11
M11
A/S 949 1689 1708 1724 2235
A/S 2409 2414 2419 2424 2429
M11
A/S 2433 2437 2441 2445 2449
A/S 2452 2481 2805 2861 2866
Y001
X001
M12
M12
A/S 1121 1728 2477 2858 2862
A/S 2896 2924 2927 2959 2962
M12
A/S 2984 2987
A/S
* Declarations
66
M8000
MOV K15 D8114
D8114
B/D 67
B/D
MOV K0 D550
D550
A/S 2615 2648 2681
B/D 72
MOV K0 D304
D304
A/S 1472 2633 2666 2699
B/D 77 2642 2675 2708
MOV K0 D303
D303
A/S 2523 2561 2599
B/D 82 2531 2569 2607
MOV K0 Z0
Z0
A/S 226 252 259
B/D 87 239
MOV K0 Z1
Z1
A/S 269 295 302
B/D 92 282
MOV K0 Z2
Z2
A/S 339 352 359
B/D 97
MOV K0 Z3
Z3
A/S 369 391 398
B/D 102 382
MOV K0 Z4
Z4
A/S 565 570 577
B/D 107
MOV K0 Z5
Z5
A/S 587 600 607
B/D 112
MOV K97 D8400
D8400
B/D 117
B/D
MOV H1 D8401
D8401
B/D 122
B/D
SET M8192
M8192
A/S 127
A/S
MOV K1 D8397
D8397
B/D 129
B/D

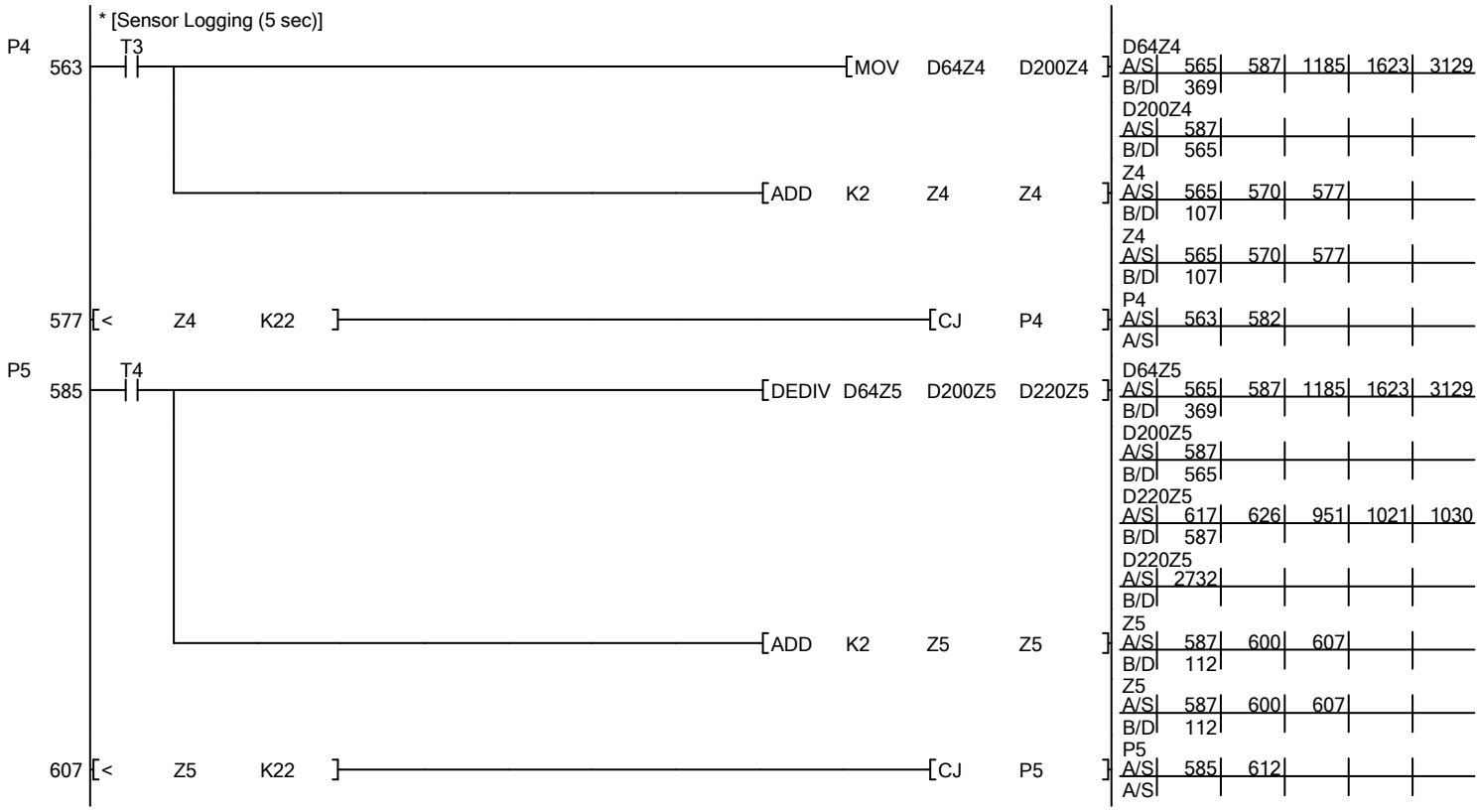

* Analog to Digital
M8000
134
MOV D8030 D0
MOV D8031 D2
MOV D8231 D20
FROM K1 K2 D102 K4
FROM K2 K2 D106 K4
D8030
A/S 135
A/S
D0
A/S 169 226
B/D 135
D8031
A/S 140
A/S
D2
A/S 174
B/D 140
D8231
A/S 145
A/S
D20
A/S 219 311
B/D 145
D102
A/S 179
B/D 150
D106
A/S 199
B/D 159

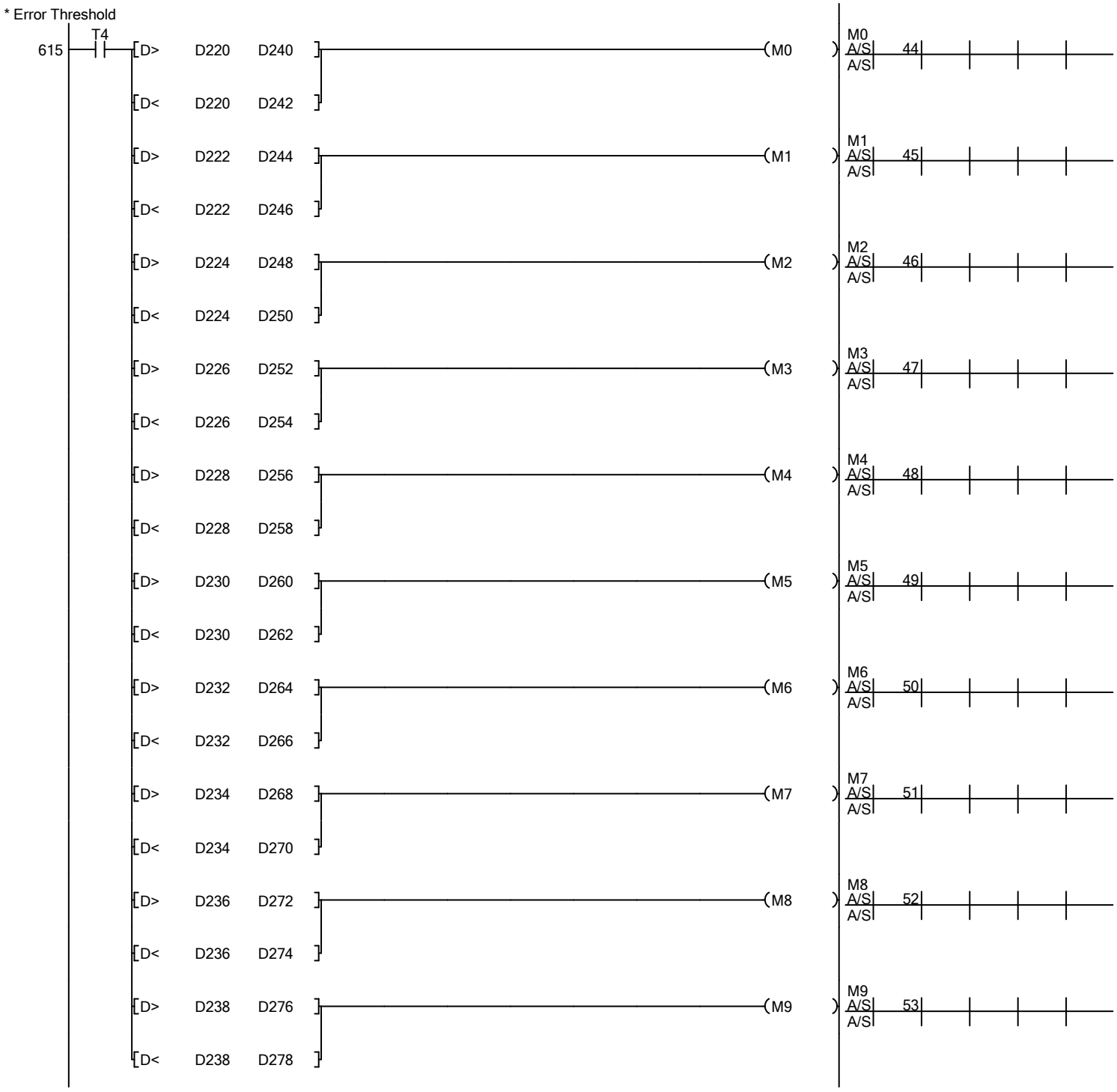

* ADC Float
M8000
168
FLT D0 D0
FLT D2 D2
FLT D102 D4
FLT D103 D6
FLT D104 D8
FLT D105 D10
FLT D106 D12
FLT D107 D14
FLT D108 D16
FLT D109 D18
FLT D20 D20
D0 A/S 169 226
B/D 135
D0 A/S 169 226
B/D 135
D2 A/S 174
B/D 140
D2 A/S 174
B/D 140
D102 A/S 179
B/D 150
D4 B/D 179
B/D
D103 A/S 184
A/S
D6 B/D 184
B/D
D104 A/S 189
A/S
D8 B/D 189
B/D
D105 A/S 194
A/S
D10 B/D 194
B/D
D106 A/S 199
B/D 159
D12 A/S 269
B/D 199
D107 A/S 204
A/S
D14 B/D 204
B/D
D108 A/S 209
A/S
D16 B/D 209
B/D
D109 A/S 214
A/S
D18 B/D 214
B/D
D20 A/S 219 311
B/D 145
D20 A/S 219 311
B/D 145


* [RTD to Temperature]
P0
T2
224
DEMUL D0Z0 E2.5 D100
DESUB D100 E1 D22Z0
ADD K2 Z0 Z0
259 < Z0 K12
CJ P0
* [Ultrasonic to Level]
P1
T2
267
DEMUL D12Z1 E2.5 D110
DESUB D110 E1 D34Z1
ADD K2 Z1 Z1
302 < Z1 K8
CJ P1
* Flowmeter to RPM
T0
310
DEMUL D20 E2.5 D112
DESUB D112 E1 D42
* [Sensor Averaging]
P2
T2
337
DEADD D22Z2 D22Z2 D44Z2
ADD K2 Z2 Z2
359 < Z2 K22
CJ P2

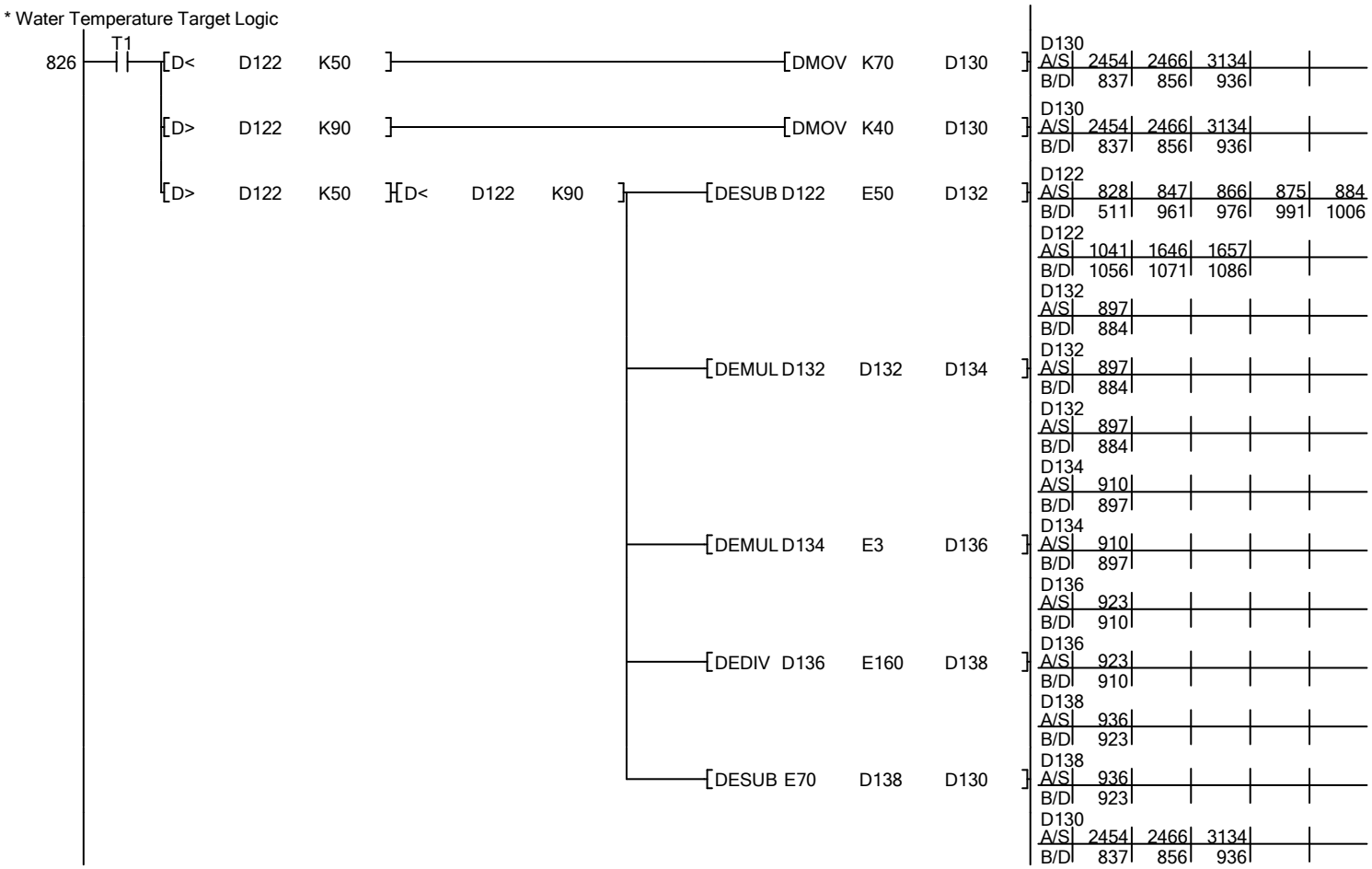

P3
T3
367
DEDIV D44Z3 E5 D64Z3
DEMOV E0 D44Z3
ADD K2 Z3 Z3
398 < Z3 K22
CJ P3
D44Z3
A/S 369
B/D 339 382
D64Z3
A/S 565 587 1185 1623 3129
B/D 369
D44Z3
A/S 369
B/D 339 382
Z3
A/S 369 391 398
B/D 102 382
Z3
A/S 369 391 398
B/D 102 382
P3
A/S 367 403
A/S

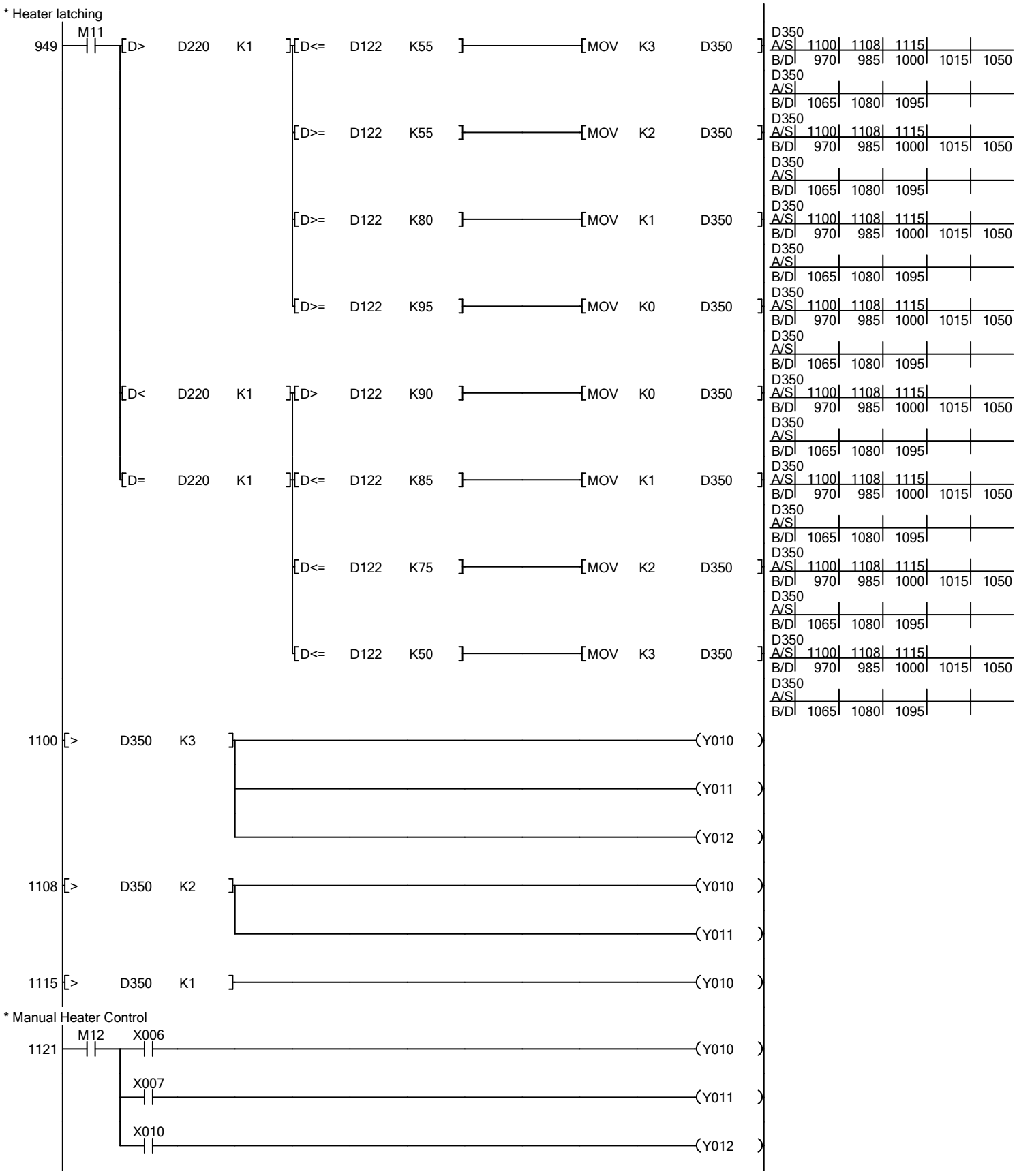

* Level % Converter
406
T1
DESUB K100 D76 D114
DESUB K100 D78 D116
DESUB K100 D80 D118
DESUB K100 D82 D120
DEDIV D114 K100 D114
DEDIV D116 K100 D116
DEDIV D118 K100 D118
DEDIV D120 K100 D120
DEMUL D114 K100 D122
DEMUL D116 K100 D124
DEMUL D118 K100 D126
DEMUL D120 K100 D128
D76 A/S 407 1132 1358 1446 3139
A/S
D114 A/S 459 511
B/D 407
D78 A/S 420 1145
A/S
D116 A/S 472 524
B/D 420
D80 A/S 433 1158
A/S
D118 A/S 485 537
B/D 433
D82 A/S 446 1171
A/S
D120 A/S 498 550
B/D 446
D114 A/S 459 511
B/D 407
D114 A/S 459 511
B/D 407
D116 A/S 472 524
B/D 420
D116 A/S 472 524
B/D 420
D118 A/S 485 537
B/D 433
D118 A/S 485 537
B/D 433
D120 A/S 498 550
B/D 446
D120 A/S 498 550
B/D 446
D114 A/S 459 511
B/D 407
D122 A/S 828 847 866 875 884
B/D 511 961 976 991 1006
D122 A/S 1041 1646 1657
B/D 1056 1071 1086
D116 A/S 472 524
B/D 420
D124 B/D 524
B/D
D118 A/S 485 537
B/D 433
D126 B/D 537
B/D
D120 A/S 498 550
B/D 446
D128 B/D 550
B/D

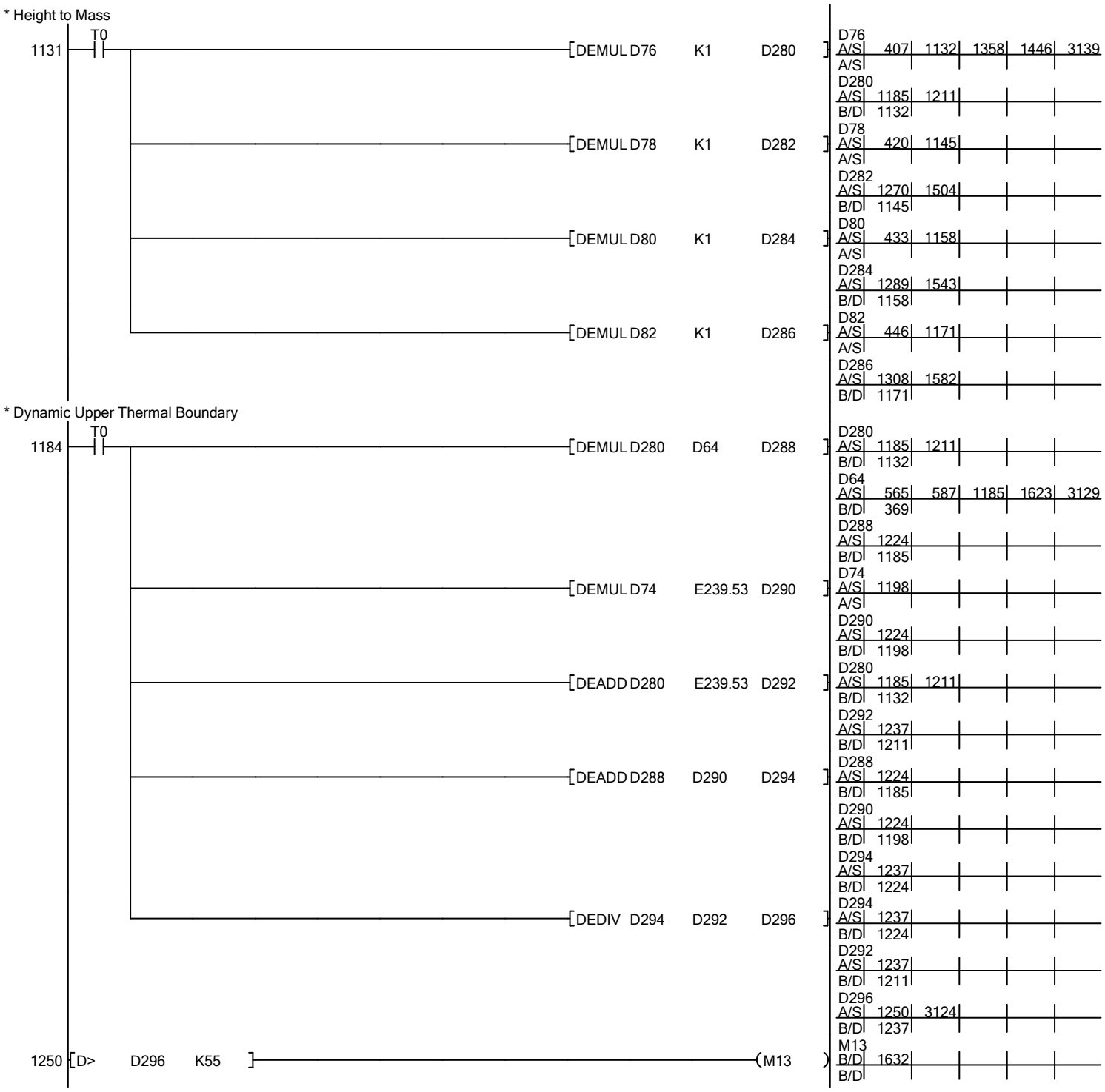

* [Sensor Logging (5 sec)]
P4
563
T3
MOV D64Z4 D200Z4
ADD K2 Z4 Z4
577
< Z4 K22
CJ P4
P5
585
T4
DEDIV D64Z5 D200Z5 D220Z5
ADD K2 Z5 Z5
607
< Z5 K22
CJ P5
D64Z4 A/S 565 587 1185 1623 3129 B/D 369
D200Z4 A/S 587 B/D 565
Z4 A/S 565 570 577 B/D 107
Z4 A/S 565 570 577 B/D 107
P4 A/S 563 582 A/S
D64Z5 A/S 565 587 1185 1623 3129 B/D 369
D200Z5 A/S 587 B/D 565
D220Z5 A/S 617 626 951 1021 1030 B/D 587
D220Z5 A/S 2732 B/D
Z5 A/S 587 600 607 B/D 112
Z5 A/S 587 600 607 B/D 112
P5 A/S 585 612 A/S

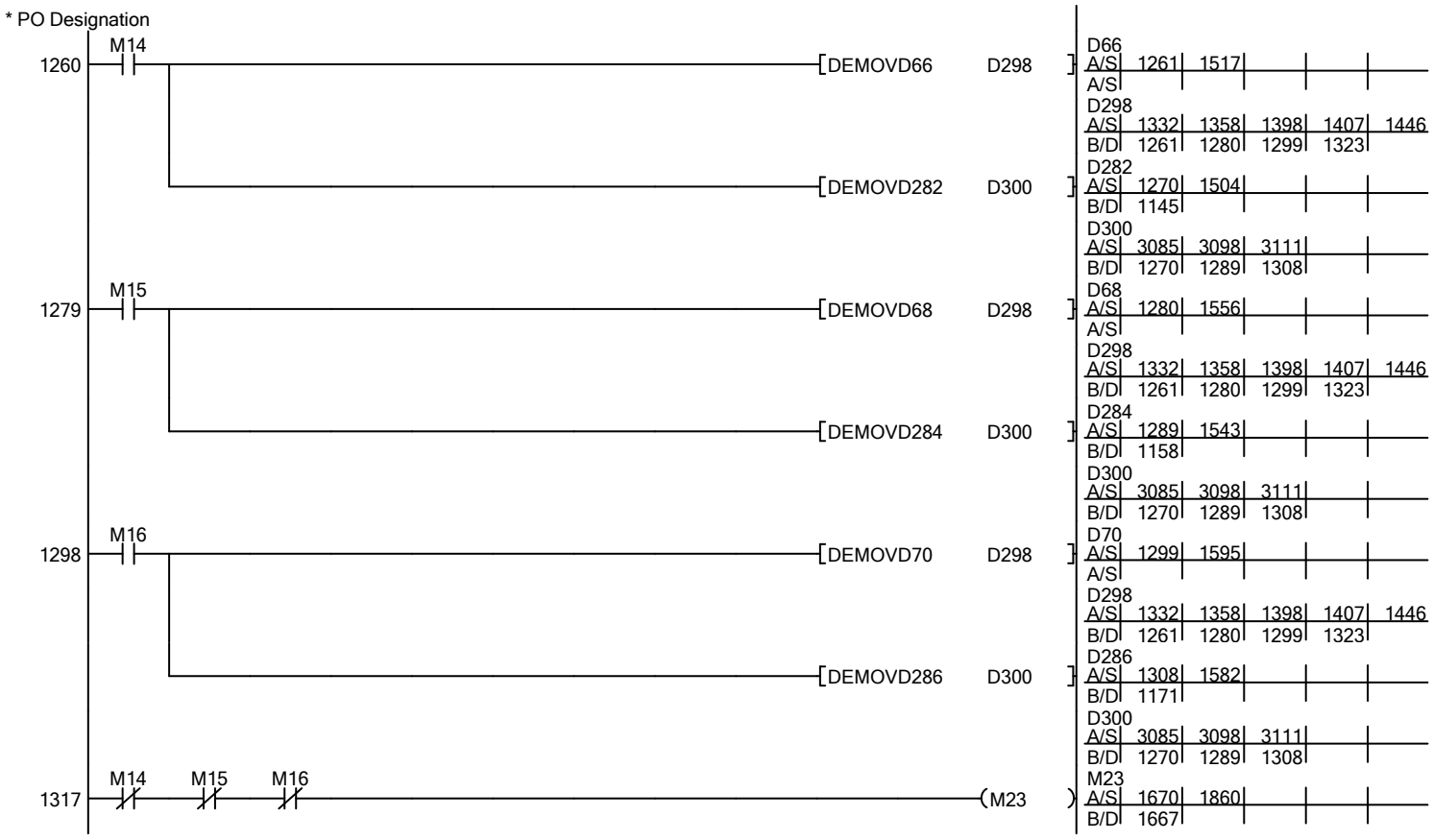

* Error Threshold
615
T4
D> D220 D240
D< D220 D242
D> D222 D244
D< D222 D246
D> D224 D248
D< D224 D250
D> D226 D252
D< D226 D254
D> D228 D256
D< D228 D258
D> D230 D260
D< D230 D262
D> D232 D264
D< D232 D266
D> D234 D268
D< D234 D270
D> D236 D272
D< D236 D274
D> D238 D276
D< D238 D278
M0
M1
M2
M3
M4
M5
M6
M7
M8
M9
M0 A/S 44 A/S
M1 A/S 45 A/S
M2 A/S 46 A/S
M3 A/S 47 A/S
M4 A/S 48 A/S
M5 A/S 49 A/S
M6 A/S 50 A/S
M7 A/S 51 A/S
M8 A/S 52 A/S
M9 A/S 53 A/S

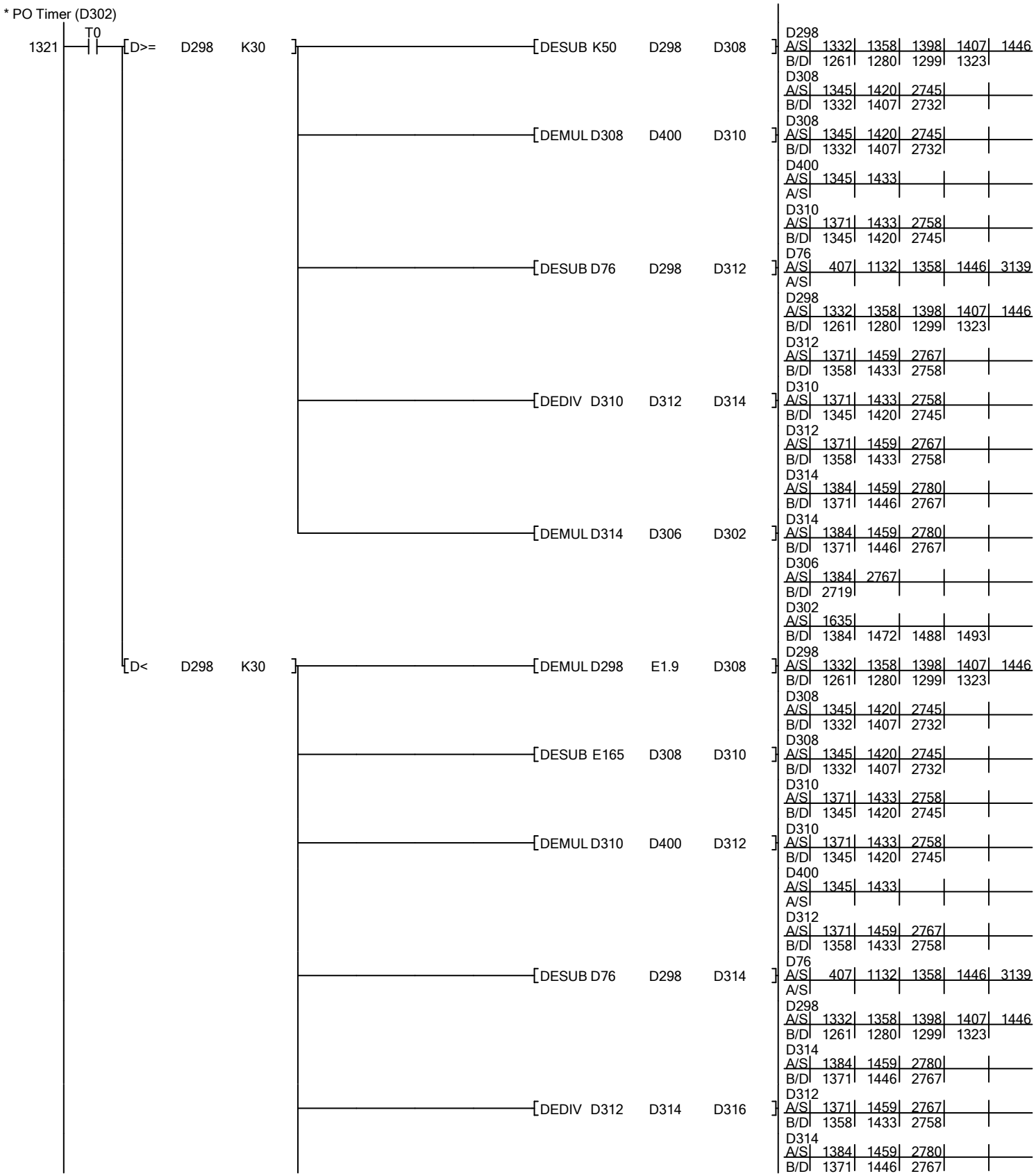
* Water Temperature Target Logic

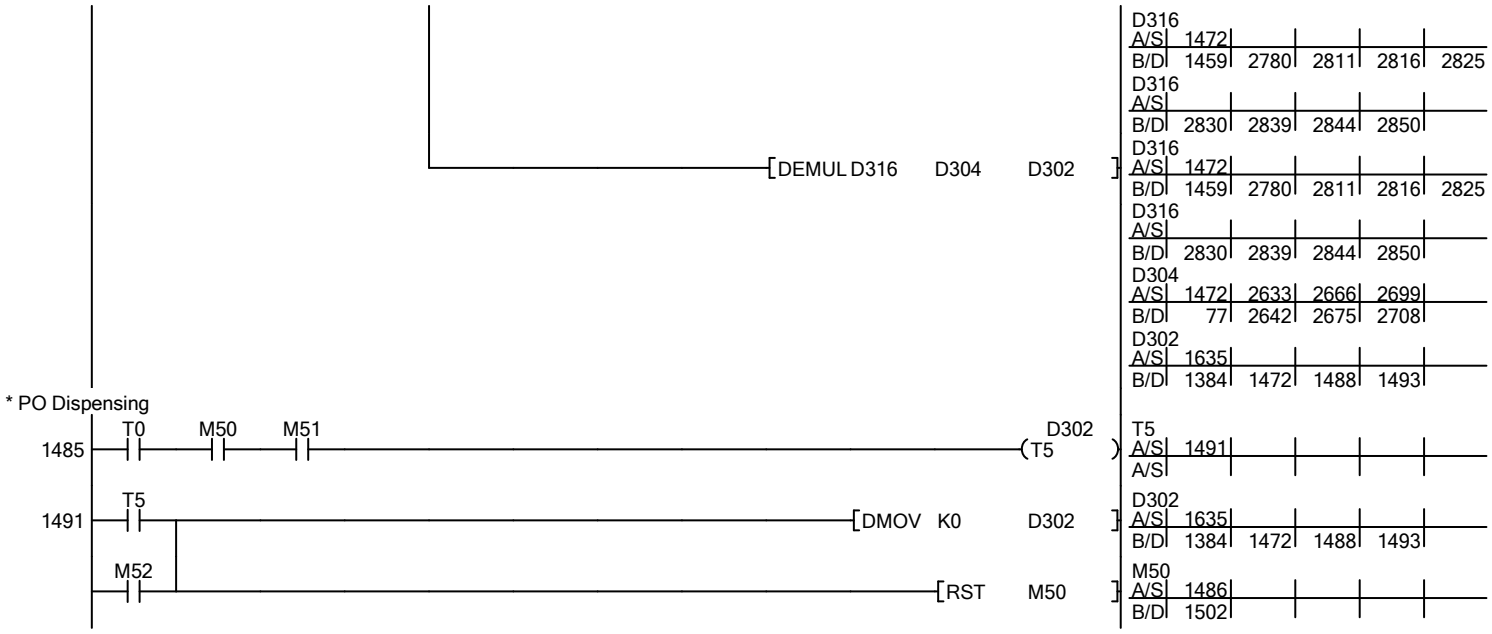


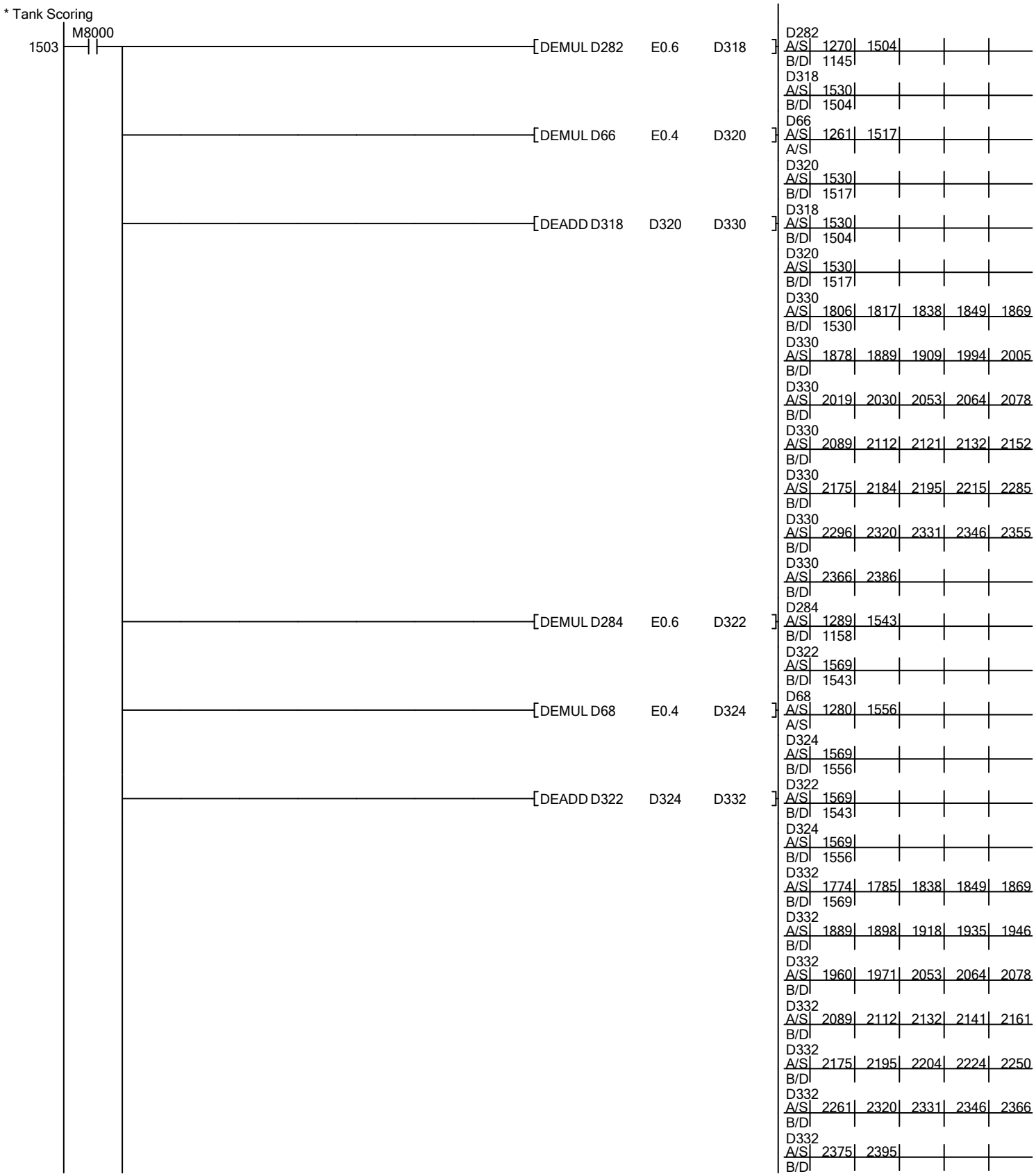

* Heater latching
949 M11 [D> D220 K1] [D<= D122 K55] [MOV K3 D350]
[D>= D122 K55] [MOV K2 D350]
[D>= D122 K80] [MOV K1 D350]
[D>= D122 K95] [MOV K0 D350]
[D< D220 K1] [D> D122 K90] [MOV K0 D350]
[D= D220 K1] [D<= D122 K85] [MOV K1 D350]
[D<= D122 K75] [MOV K2 D350]
[D<= D122 K50] [MOV K3 D350]
D350 A/S 1100 1108 1115
B/D 970 985 1000 1015 1050
D350 A/S
B/D 1065 1080 1095
1100 [> D350 K3] (Y010) (Y011) (Y012)
1108 [> D350 K2] (Y010) (Y011)
1115 [> D350 K1] (Y010)
* Manual Heater Control
1121 M12 X006 (Y010)
X007 (Y011)
X010 (Y012)

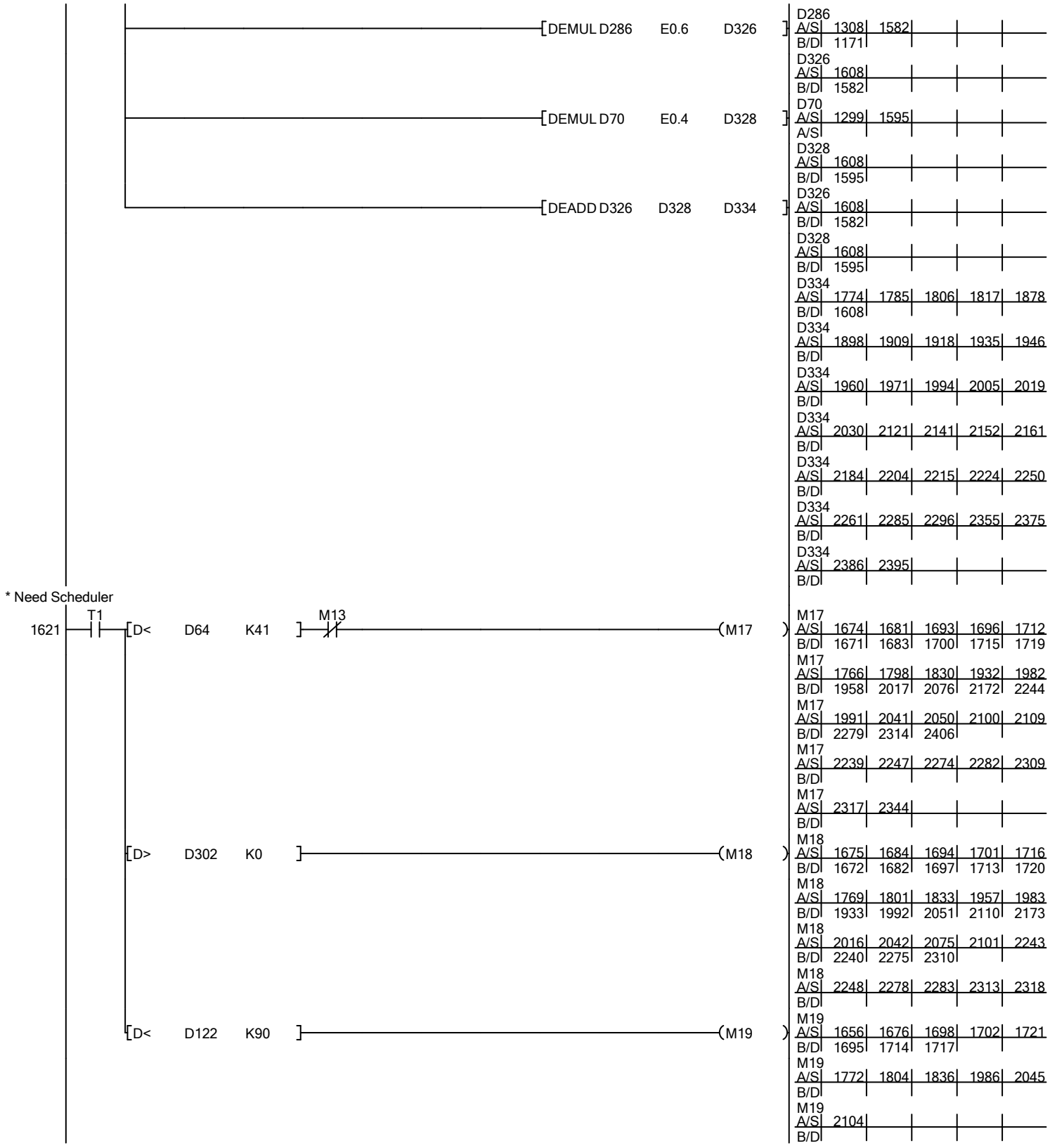

* Height to Mass
1131 T0
DEMUL D76 K1 D280
DEMUL D78 K1 D282
DEMUL D80 K1 D284
DEMUL D82 K1 D286
D76 A/S 407 1132 1358 1446 3139
D280 A/S 1185 1211 B/D 1132
D78 A/S 420 1145
D282 A/S 1270 1504 B/D 1145
D80 A/S 433 1158
D284 A/S 1289 1543 B/D 1158
D82 A/S 446 1171
D286 A/S 1308 1582 B/D 1171
* Dynamic Upper Thermal Boundary
1184 T0
DEMUL D280 D64 D288
DEMUL D74 E239.53 D290
DEADD D280 E239.53 D292
DEADD D288 D290 D294
DEDIV D294 D292 D296
D64 A/S 565 587 1185 1623 3129 B/D 369
D74 A/S 1198
D288 A/S 1224 B/D 1185
D290 A/S 1224 B/D 1198
D292 A/S 1237 B/D 1211
D294 A/S 1237 B/D 1224
D296 A/S 1250 3124 B/D 1237
1250 D> D296 K55
M13
M13 B/D 1632

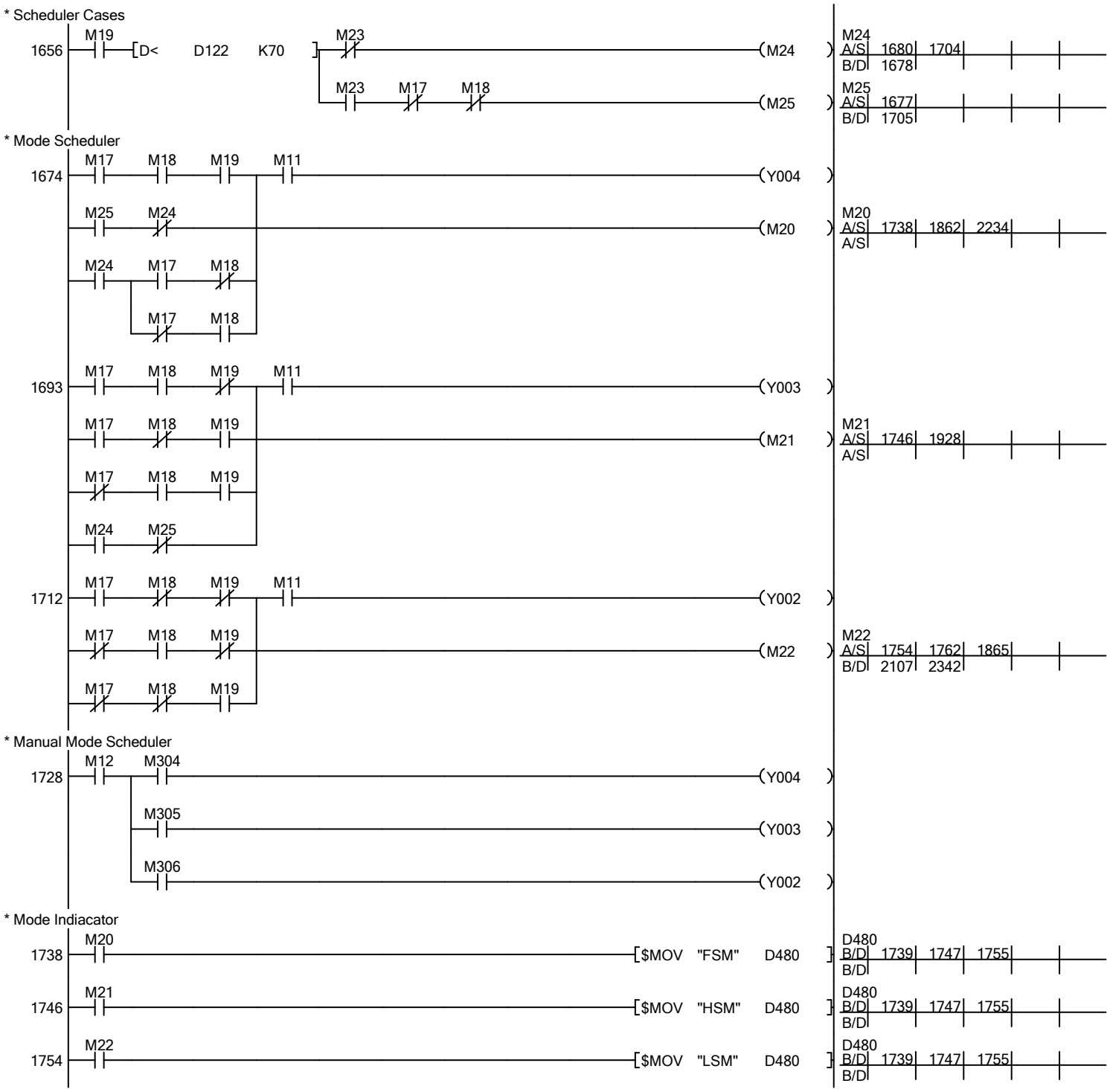

* PO Designation
1260
M14
DEMOVD66 D298
DEMOVD282 D300
1279
M15
DEMOVD68 D298
DEMOVD284 D300
1298
M16
DEMOVD70 D298
DEMOVD286 D300
1317
M14 M15 M16
M23
D66 A/S 1261 1517 A/S
D298 A/S 1332 1358 1398 1407 1446 B/D 1261 1280 1299 1323
D282 A/S 1270 1504 B/D 1145
D300 A/S 3085 3098 3111 B/D 1270 1289 1308
D68 A/S 1280 1556 A/S
D298 A/S 1332 1358 1398 1407 1446 B/D 1261 1280 1299 1323
D284 A/S 1289 1543 B/D 1158
D300 A/S 3085 3098 3111 B/D 1270 1289 1308
D70 A/S 1299 1595 A/S
D298 A/S 1332 1358 1398 1407 1446 B/D 1261 1280 1299 1323
D286 A/S 1308 1582 B/D 1171
D300 A/S 3085 3098 3111 B/D 1270 1289 1308
M23 A/S 1670 1860 B/D 1667

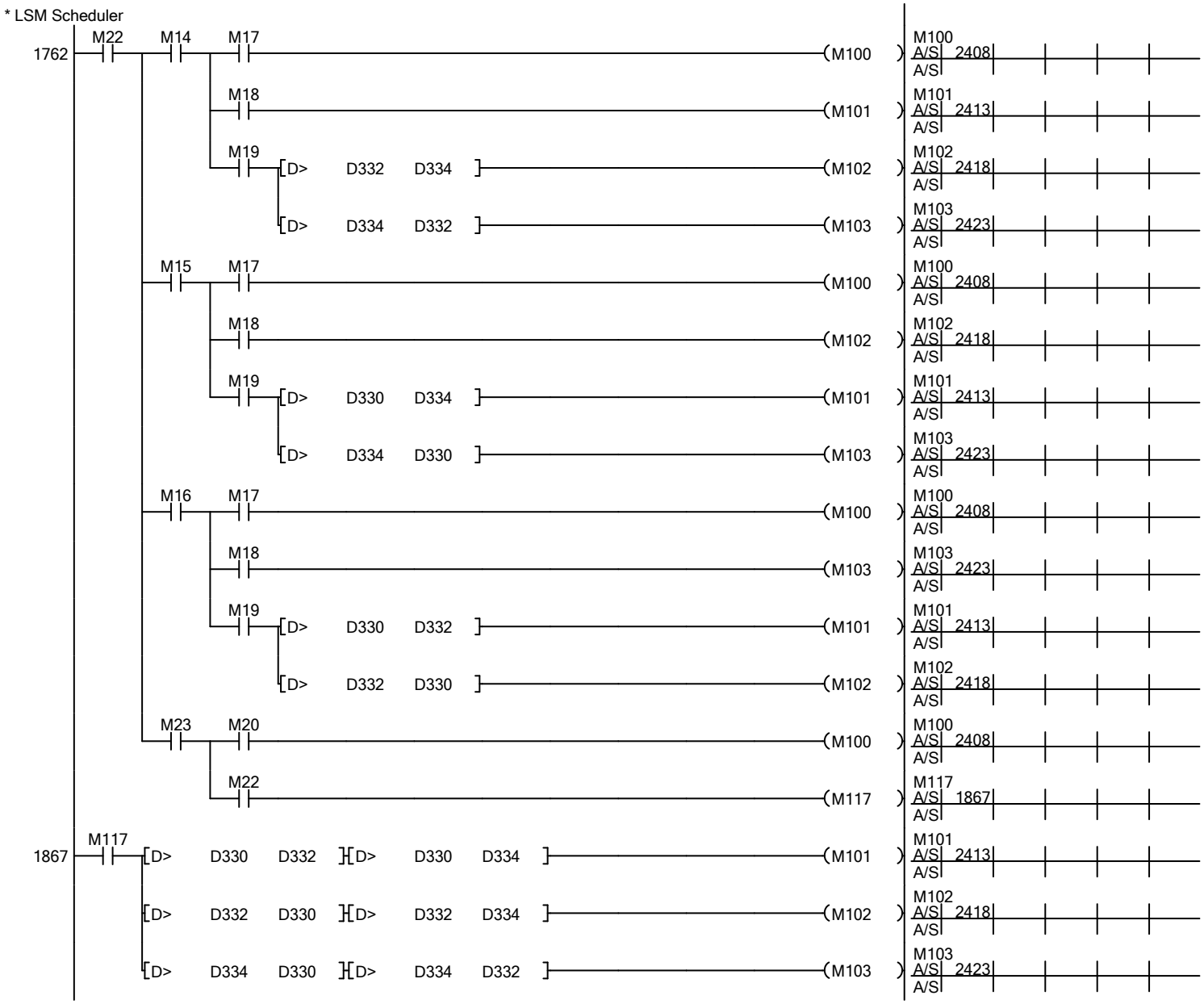

* PO Timer (D302)
1321
T0
D>= D298 K30
D< D298 K30
DESUB K50 D298 D308
DEMUL D308 D400 D310
DESUB D76 D298 D312
DEDIV D310 D312 D314
DEMUL D314 D306 D302
DEMUL D298 E1.9 D308
DESUB E165 D308 D310
DEMUL D310 D400 D312
DESUB D76 D298 D314
DEDIV D312 D314 D316

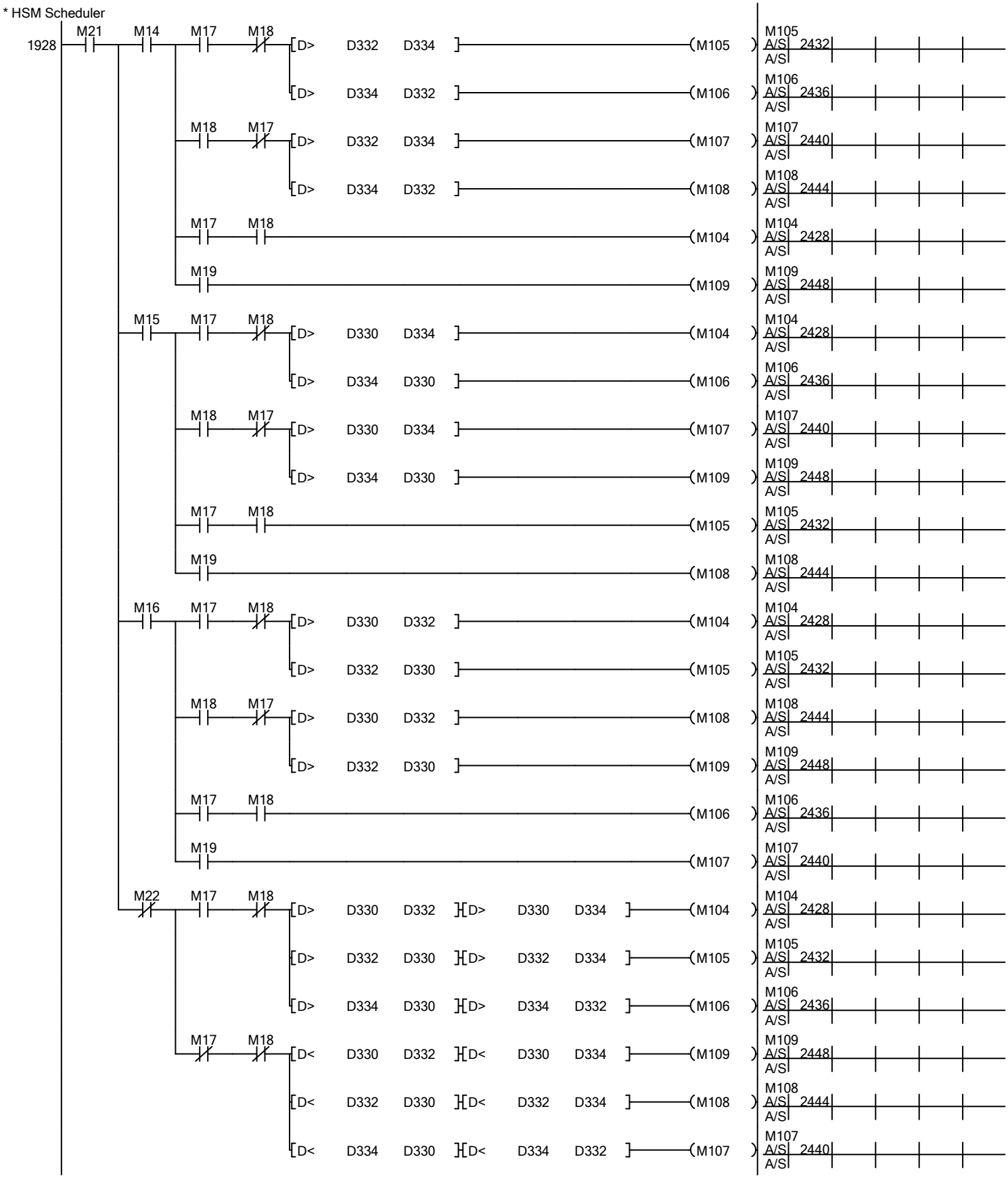

[DEMUL D316 D304 D302]
D316 A/S 1472 B/D 1459 2780 2811 2816 2825
D316 A/S B/D 2830 2839 2844 2850
D316 A/S 1472 B/D 1459 2780 2811 2816 2825
D316 A/S B/D 2830 2839 2844 2850
D304 A/S 1472 2633 2666 2699 B/D 77 2642 2675 2708
D302 A/S 1635 B/D 1384 1472 1488 1493
* PO Dispensing
1485 T0 M50 M51 (T5 D302)
T5 A/S 1491 A/S
1491 T5 [DMOV K0 D302]
D302 A/S 1635 B/D 1384 1472 1488 1493
M52 [RST M50]
M50 A/S 1486 B/D 1502

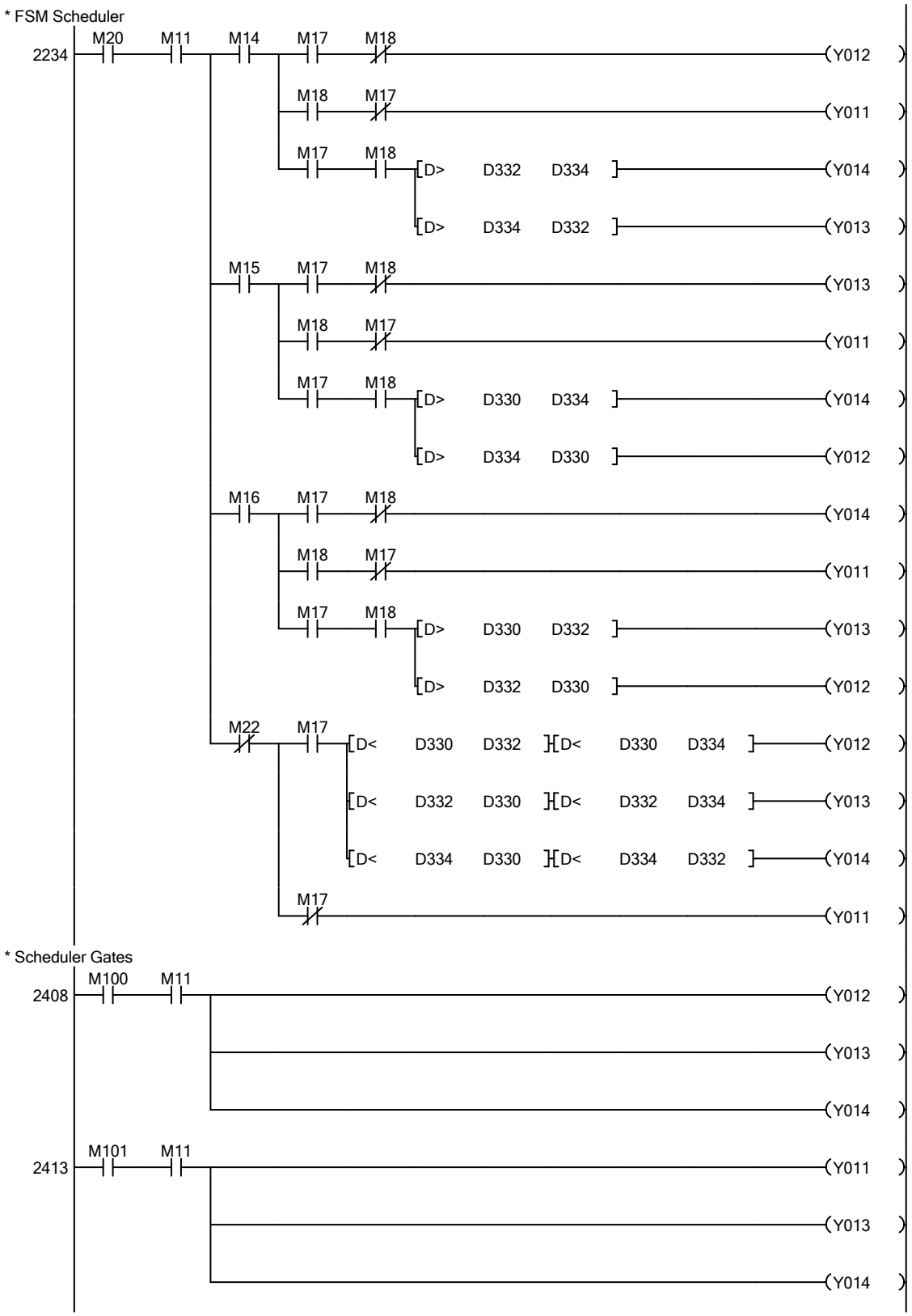

* Tank Scoring
1503
M8000
DEMUL D282 E0.6 D318
D282 A/S 1270 1504 B/D 1145
D318 A/S 1530 B/D 1504
DEMUL D66 E0.4 D320
D66 A/S 1261 1517 A/S
D320 A/S 1530 B/D 1517
DEADD D318 D320 D330
D318 A/S 1530 B/D 1504
D320 A/S 1530 B/D 1517
D330 A/S 1806 1817 1838 1849 1869 B/D 1530
D330 A/S 1878 1889 1909 1994 2005 B/D
D330 A/S 2019 2030 2053 2064 2078 B/D
D330 A/S 2089 2112 2121 2132 2152 B/D
D330 A/S 2175 2184 2195 2215 2285 B/D
D330 A/S 2296 2320 2331 2346 2355 B/D
D330 A/S 2366 2386 B/D
DEMUL D284 E0.6 D322
D284 A/S 1289 1543 B/D 1158
D322 A/S 1569 B/D 1543
DEMUL D68 E0.4 D324
D68 A/S 1280 1556 A/S
D324 A/S 1569 B/D 1556
DEADD D322 D324 D332
D322 A/S 1569 B/D 1543
D324 A/S 1569 B/D 1556
D332 A/S 1774 1785 1838 1849 1869 B/D 1569
D332 A/S 1889 1898 1918 1935 1946 B/D
D332 A/S 1960 1971 2053 2064 2078 B/D
D332 A/S 2089 2112 2132 2141 2161 B/D
D332 A/S 2175 2195 2204 2224 2250 B/D
D332 A/S 2261 2320 2331 2346 2366 B/D
D332 A/S 2375 2395 B/D

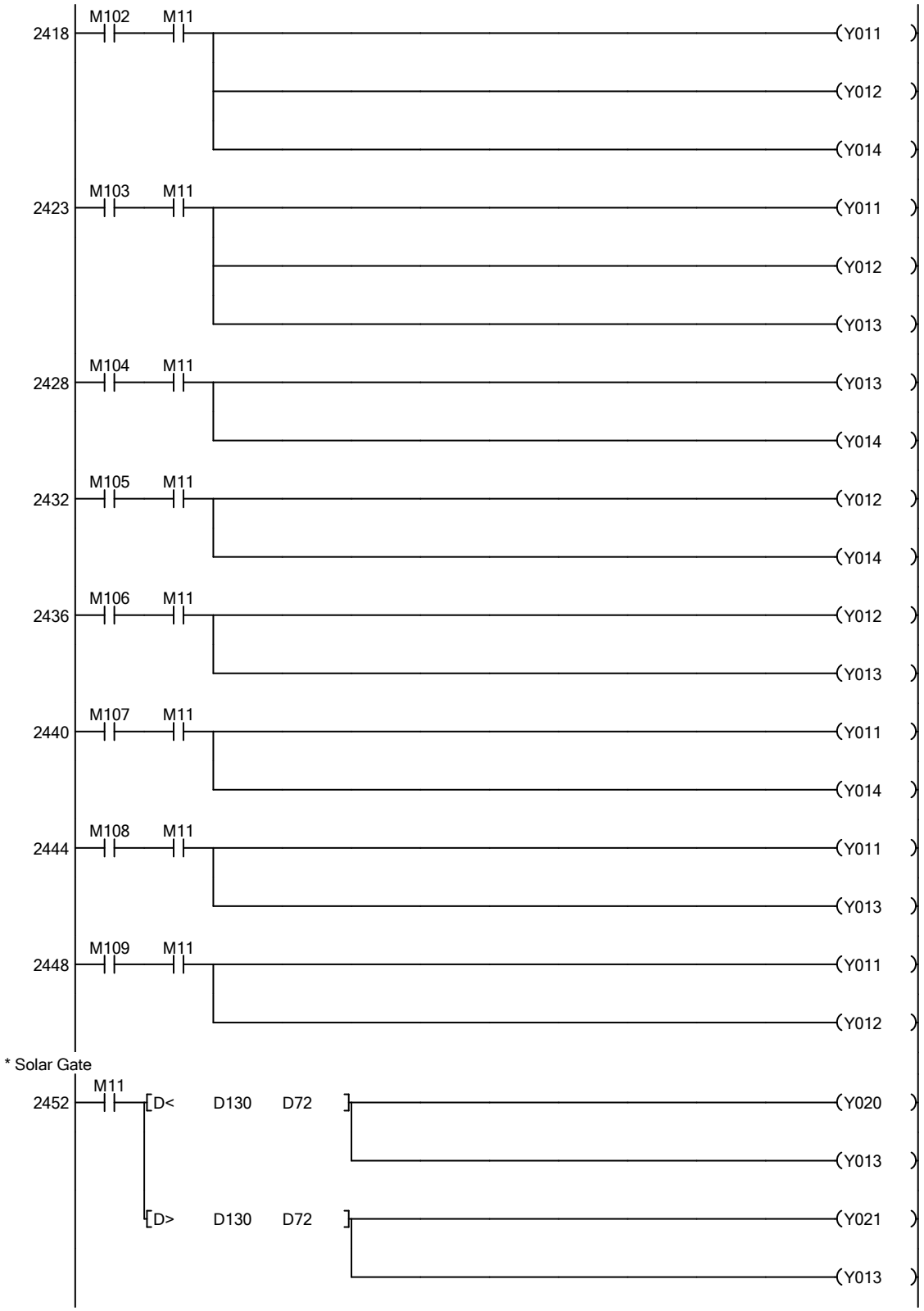

[DEMUL D286 E0.6 D326]
[DEMUL D70 E0.4 D328]
[DEADD D326 D328 D334]
* Need Scheduler
1621
T1
[D< D64 K41]
M13
(M17)
[D> D302 K0]
(M18)
[D< D122 K90]
(M19)

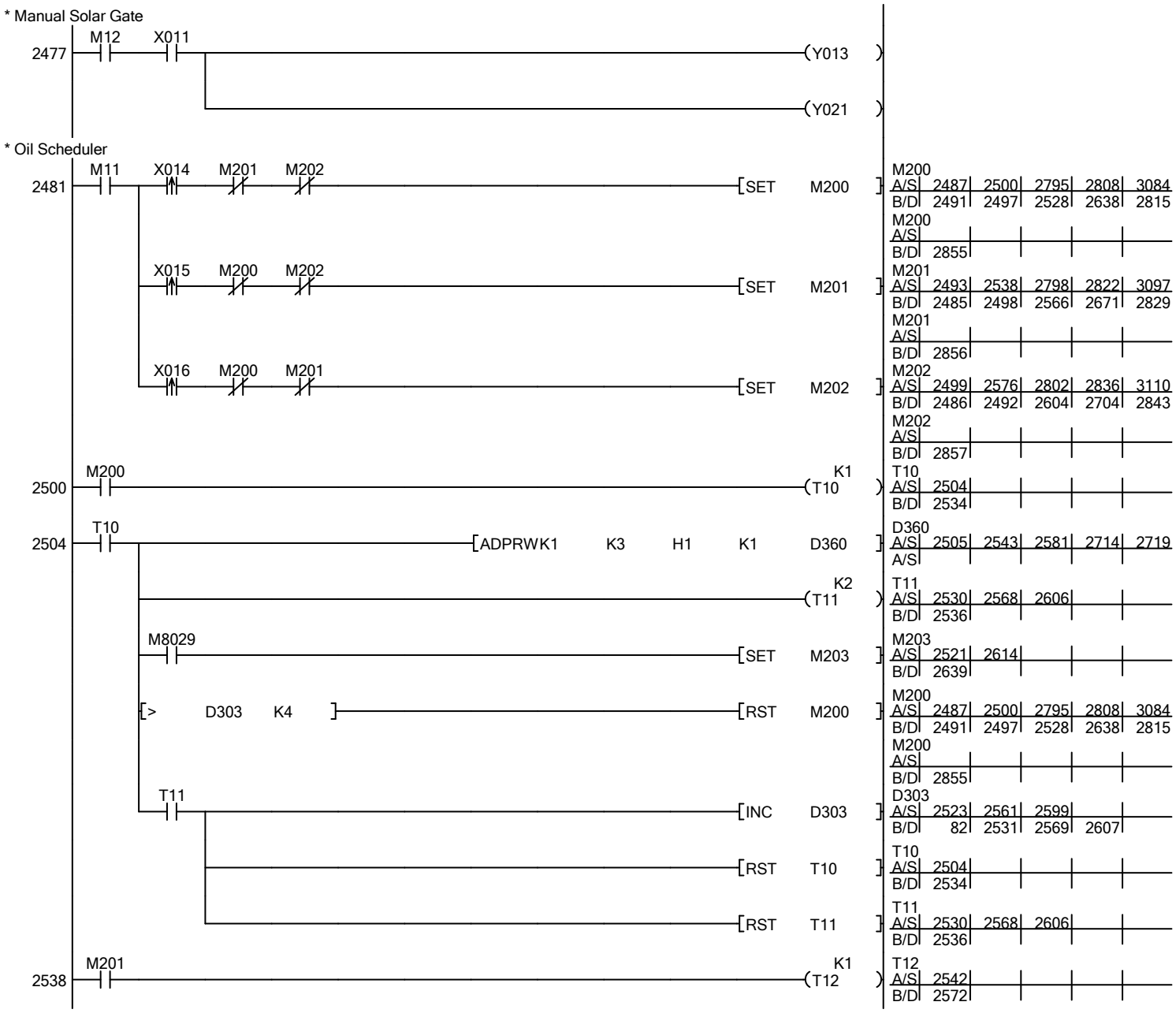

* Scheduler Cases
1656
M19
D< D122 K70
M23
M24
M23 M17 M18
M25
M24 A/S 1680 1704
B/D 1678
M25 A/S 1677
B/D 1705
* Mode Scheduler
1674
M17 M18 M19 M11
Y004
M25 M24
M20
M24 M17 M18
M17 M18
M20 A/S 1738 1862 2234
A/S
1693
M17 M18 M19 M11
Y003
M17 M18 M19
M21
M17 M18 M19
M24 M25
M21 A/S 1746 1928
A/S
1712
M17 M18 M19 M11
Y002
M17 M18 M19
M22
M17 M18 M19
M22 A/S 1754 1762 1865
B/D 2107 2342
* Manual Mode Scheduler
1728
M12 M304
Y004
M305
Y003
M306
Y002
* Mode Indiacator
1738
M20
$MOV "FSM" D480
1746
M21
$MOV "HSM" D480
1754
M22
$MOV "LSM" D480
D480 B/D 1739 1747 1755
B/D

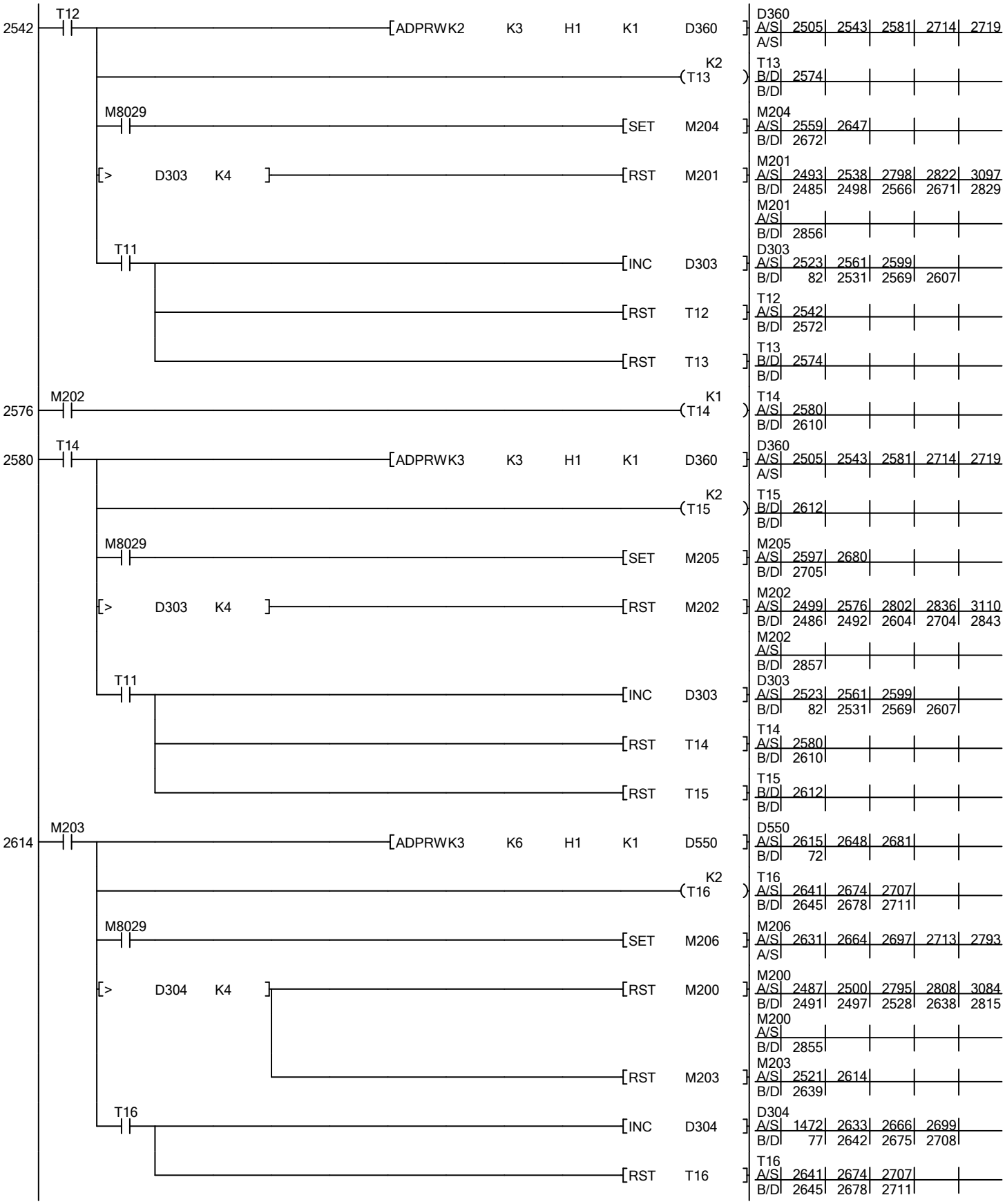

* LSM Scheduler
1762
M22
M14
M17
M100
M100 A/S 2408 A/S
M18
M101
M101 A/S 2413 A/S
M19
D> D332 D334
M102
M102 A/S 2418 A/S
D> D334 D332
M103
M103 A/S 2423 A/S
M15
M17
M100
M100 A/S 2408 A/S
M18
M102
M102 A/S 2418 A/S
M19
D> D330 D334
M101
M101 A/S 2413 A/S
D> D334 D330
M103
M103 A/S 2423 A/S
M16
M17
M100
M100 A/S 2408 A/S
M18
M103
M103 A/S 2423 A/S
M19
D> D330 D332
M101
M101 A/S 2413 A/S
D> D332 D330
M102
M102 A/S 2418 A/S
M23
M20
M100
M100 A/S 2408 A/S
M22
M117
M117 A/S 1867 A/S
1867
M117
D> D330 D332
D> D330 D334
M101
M101 A/S 2413 A/S
D> D332 D330
D> D332 D334
M102
M102 A/S 2418 A/S
D> D334 D330
D> D334 D332
M103
M103 A/S 2423 A/S

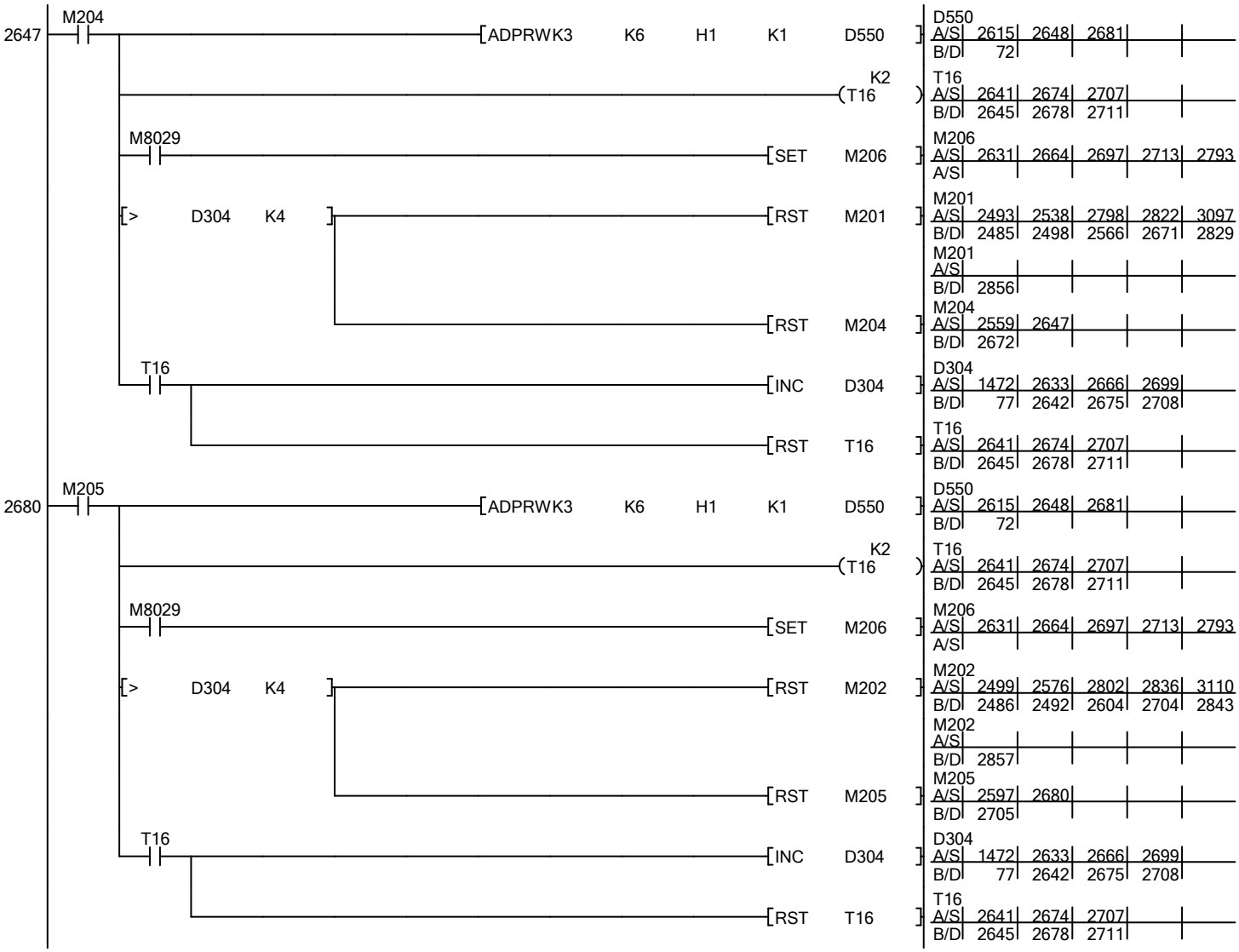

* HSM Scheduler
1928
M21 M14 M17 M18 [D> D332 D334] (M105)
[D> D334 D332] (M106)
M18 M17 [D> D332 D334] (M107)
[D> D334 D332] (M108)
M17 M18 (M104)
M19 (M109)
M15 M17 M18 [D> D330 D334] (M104)
[D> D334 D330] (M106)
M18 M17 [D> D330 D334] (M107)
[D> D334 D330] (M109)
M17 M18 (M105)
M19 (M108)
M16 M17 M18 [D> D330 D332] (M104)
[D> D332 D330] (M105)
M18 M17 [D> D330 D332] (M108)
[D> D332 D330] (M109)
M17 M18 (M106)
M19 (M107)
M22 M17 M18 [D> D330 D332][D> D330 D334] (M104)
[D> D332 D330][D> D332 D334] (M105)
[D> D334 D330][D> D334 D332] (M106)
M17 M18 [D< D330 D332][D< D330 D334] (M109)
[D< D332 D330][D< D332 D334] (M108)
[D< D334 D330][D< D334 D332] (M107)
M105 A/S 2432 A/S
M106 A/S 2436 A/S
M107 A/S 2440 A/S
M108 A/S 2444 A/S
M104 A/S 2428 A/S
M109 A/S 2448 A/S

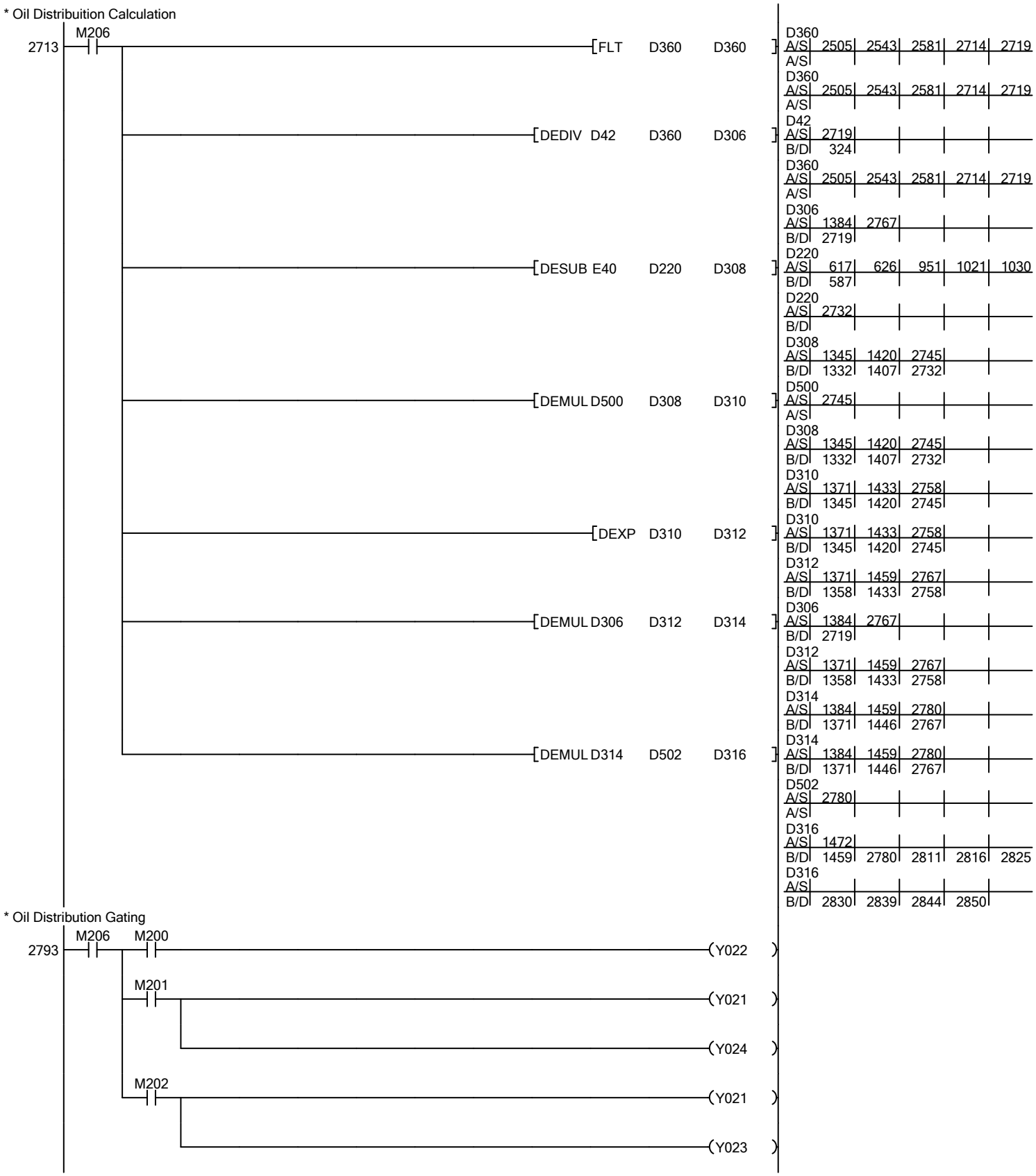

* FSM Scheduler
2234
M20 M11 M14 M17 M18 (Y012)
M18 M17 (Y011)
M17 M18 [D> D332 D334] (Y014)
[D> D334 D332] (Y013)
M15 M17 M18 (Y013)
M18 M17 (Y011)
M17 M18 [D> D330 D334] (Y014)
[D> D334 D330] (Y012)
M16 M17 M18 (Y014)
M18 M17 (Y011)
M17 M18 [D> D330 D332] (Y013)
[D> D332 D330] (Y012)
M22 M17 [D< D330 D332][D< D330 D334] (Y012)
[D< D332 D330][D< D332 D334] (Y013)
[D< D334 D330][D< D334 D332] (Y014)
M17 (Y011)
* Scheduler Gates
2408
M100 M11 (Y012)
(Y013)
(Y014)
2413
M101 M11 (Y011)
(Y013)
(Y014)

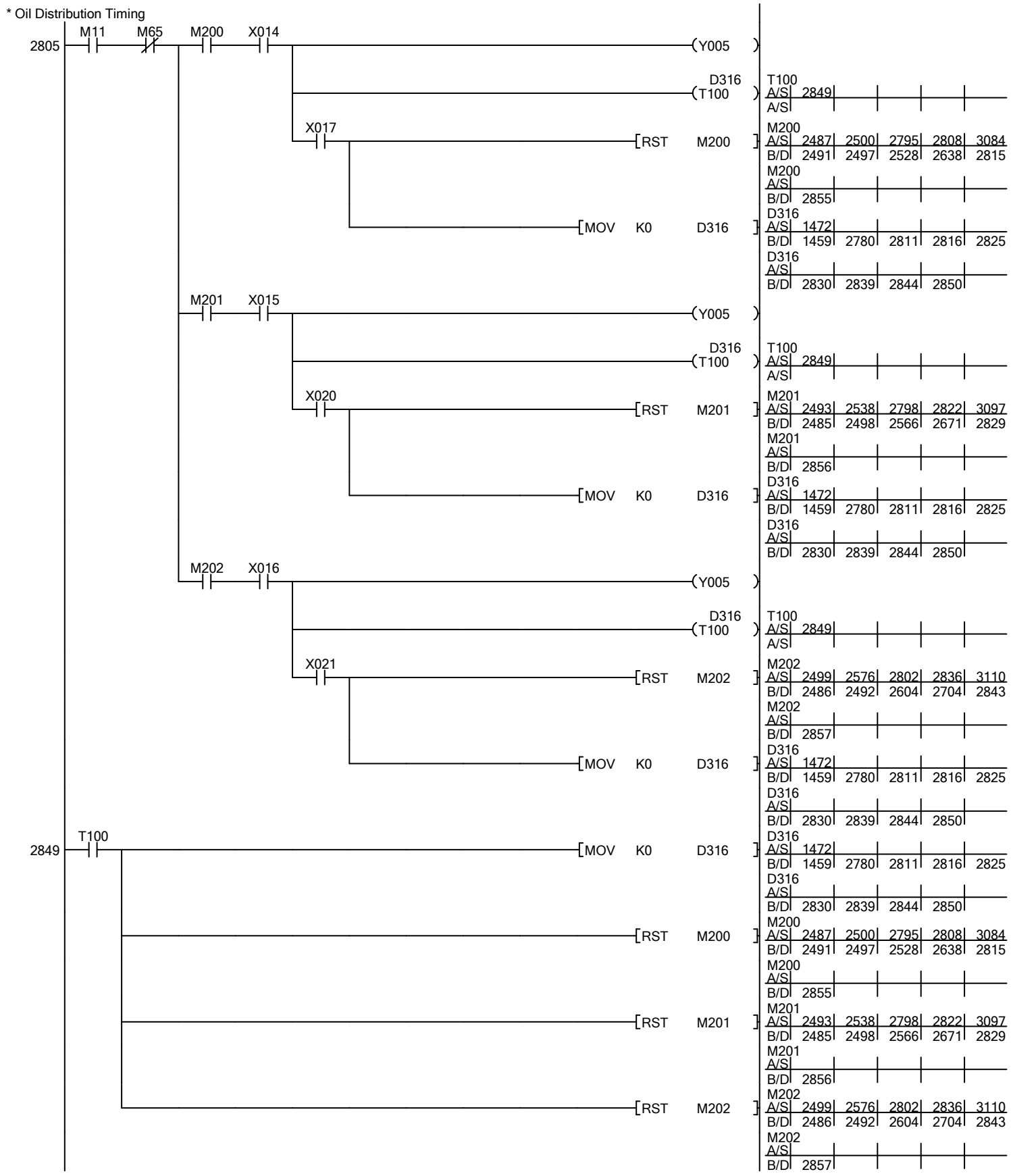

2418
M102
M11
Y011
Y012
Y014
2423
M103
M11
Y011
Y012
Y013
2428
M104
M11
Y013
Y014
2432
M105
M11
Y012
Y014
2436
M106
M11
Y012
Y013
2440
M107
M11
Y011
Y014
2444
M108
M11
Y011
Y013
2448
M109
M11
Y011
Y012
* Solar Gate
2452
M11
D< D130 D72
Y020
Y013
D> D130 D72
Y021
Y013

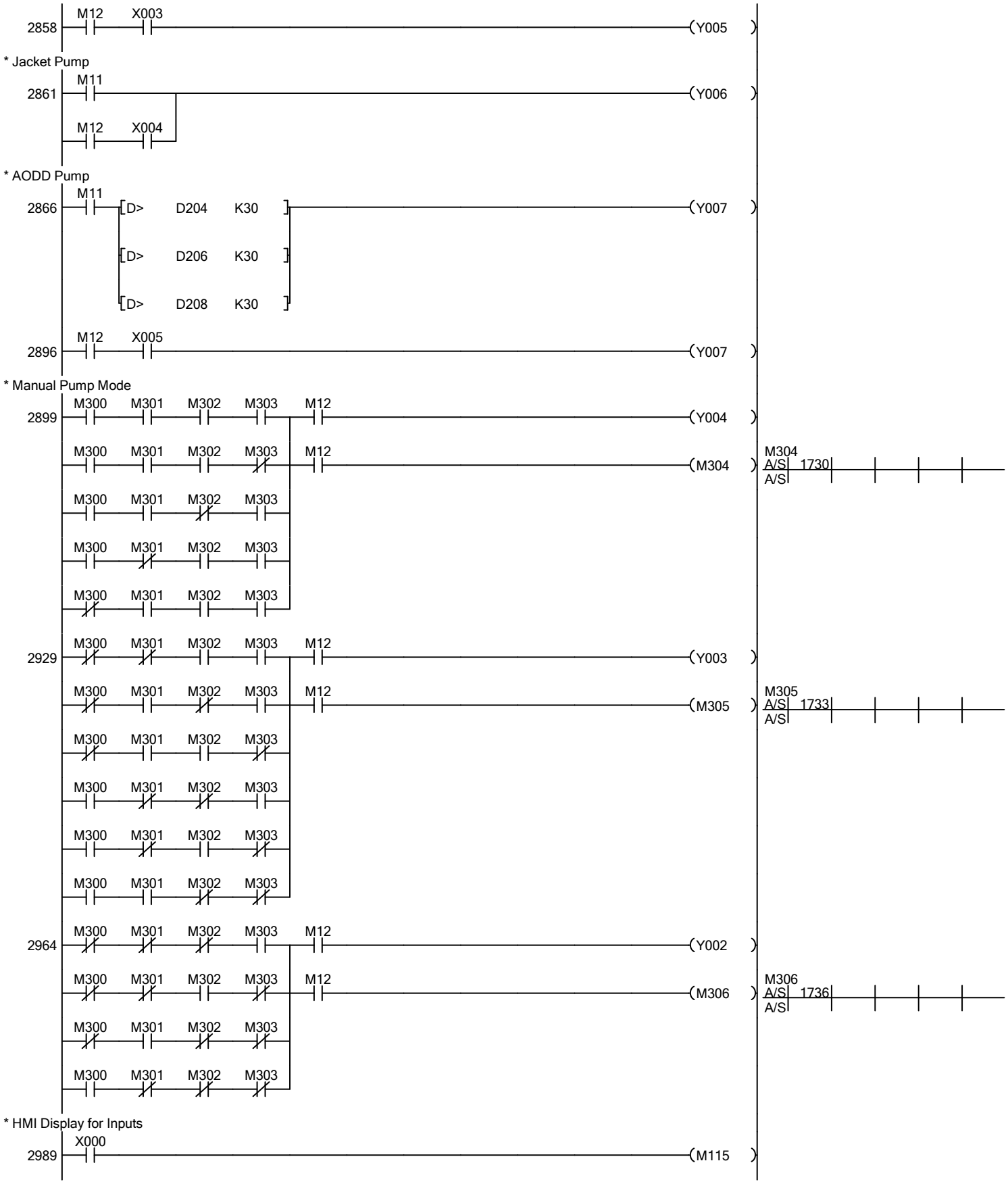

* Manual Solar Gate
2477
M12
X011
Y013
Y021
* Oil Scheduler
2481
M11
X014
M201
M202
SET M200
X015
M200
M202
SET M201
X016
M200
M201
SET M202
2500
M200
K1
T10
2504
T10
ADPRWK1 K3 H1 K1 D360
K2
T11
M8029
SET M203
> D303 K4
RST M200
T11
INC D303
RST T10
RST T11
2538
M201
K1
T12
M200 A/S 2487 2500 2795 2808 3084 B/D 2491 2497 2528 2638 2815
M200 A/S B/D 2855
M201 A/S 2493 2538 2798 2822 3097 B/D 2485 2498 2566 2671 2829
M201 A/S B/D 2856
M202 A/S 2499 2576 2802 2836 3110 B/D 2486 2492 2604 2704 2843
M202 A/S B/D 2857
T10 A/S 2504 B/D 2534
D360 A/S 2505 2543 2581 2714 2719 A/S
T11 A/S 2530 2568 2606 B/D 2536
M203 A/S 2521 2614 B/D 2639
M200 A/S 2487 2500 2795 2808 3084 B/D 2491 2497 2528 2638 2815
M200 A/S B/D 2855
D303 A/S 2523 2561 2599 B/D 82 2531 2569 2607
T10 A/S 2504 B/D 2534
T11 A/S 2530 2568 2606 B/D 2536
T12 A/S 2542 B/D 2572

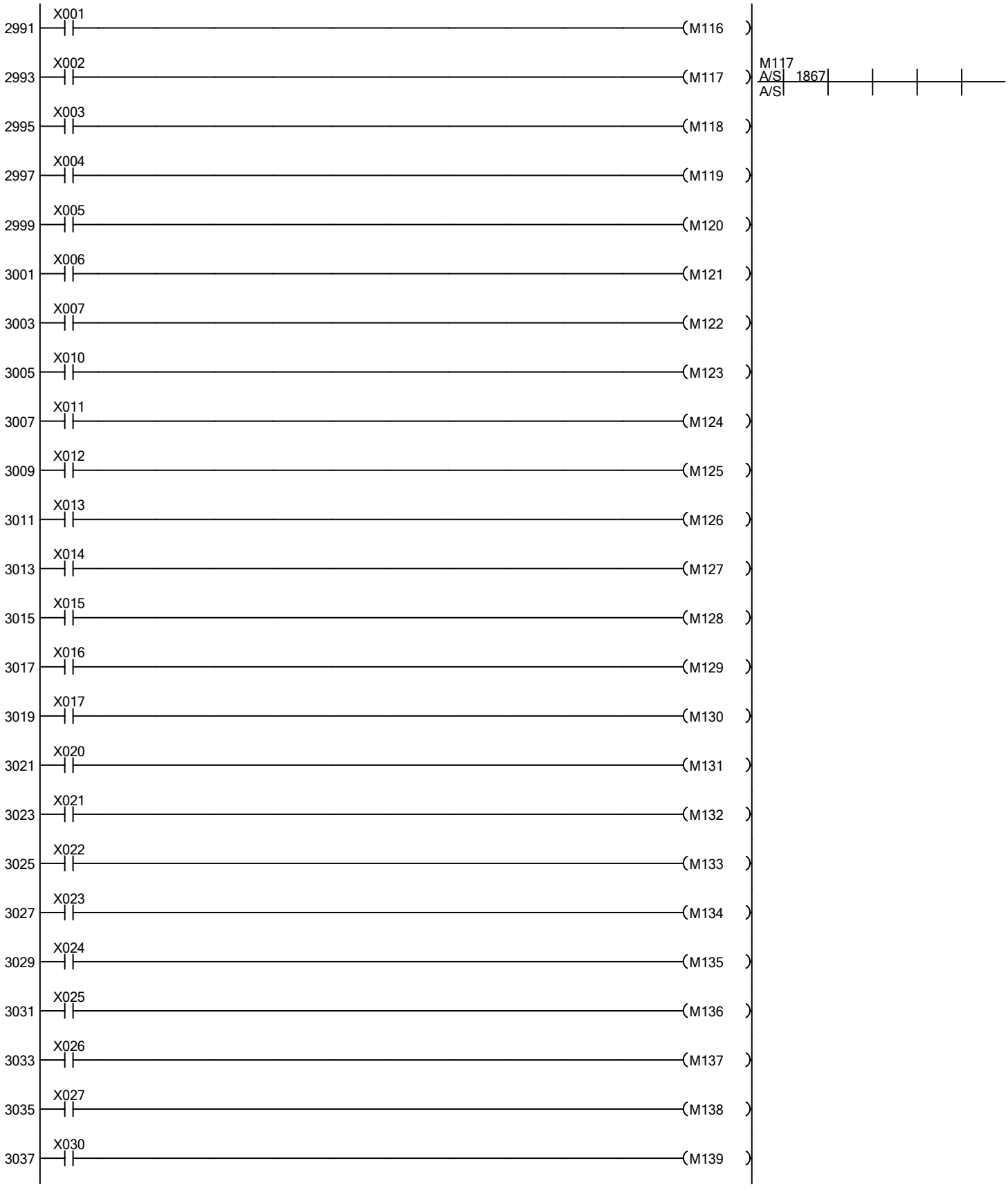

2542 T12 [ADPRW K2 K3 H1 K1 D360]
(T13 K2)
M8029 [SET M204]
[> D303 K4] [RST M201]
T11 [INC D303]
[RST T12]
[RST T13]
2576 M202 (T14 K1)
2580 T14 [ADPRW K3 K3 H1 K1 D360]
(T15 K2)
M8029 [SET M205]
[> D303 K4] [RST M202]
T11 [INC D303]
[RST T14]
[RST T15]
2614 M203 [ADPRW K3 K6 H1 K1 D550]
(T16 K2)
M8029 [SET M206]
[> D304 K4] [RST M200]
[RST M203]
T16 [INC D304]
[RST T16]
D360 A/S 2505 2543 2581 2714 2719
T13 B/D 2574
M204 A/S 2559 2647 B/D 2672
M201 A/S 2493 2538 2798 2822 3097 B/D 2485 2498 2566 2671 2829
M201 A/S B/D 2856
D303 A/S 2523 2561 2599 B/D 82 2531 2569 2607
T12 A/S 2542 B/D 2572
T13 B/D 2574
T14 A/S 2580 B/D 2610
D360 A/S 2505 2543 2581 2714 2719
T15 B/D 2612
M205 A/S 2597 2680 B/D 2705
M202 A/S 2499 2576 2802 2836 3110 B/D 2486 2492 2604 2704 2843
M202 A/S B/D 2857
D303 A/S 2523 2561 2599 B/D 82 2531 2569 2607
T14 A/S 2580 B/D 2610
T15 B/D 2612
D550 A/S 2615 2648 2681 B/D 72
T16 A/S 2641 2674 2707 B/D 2645 2678 2711
M206 A/S 2631 2664 2697 2713 2793
M200 A/S 2487 2500 2795 2808 3084 B/D 2491 2497 2528 2638 2815
M200 A/S B/D 2855
M203 A/S 2521 2614 B/D 2639
D304 A/S 1472 2633 2666 2699 B/D 77 2642 2675 2708
T16 A/S 2641 2674 2707 B/D 2645 2678 2711

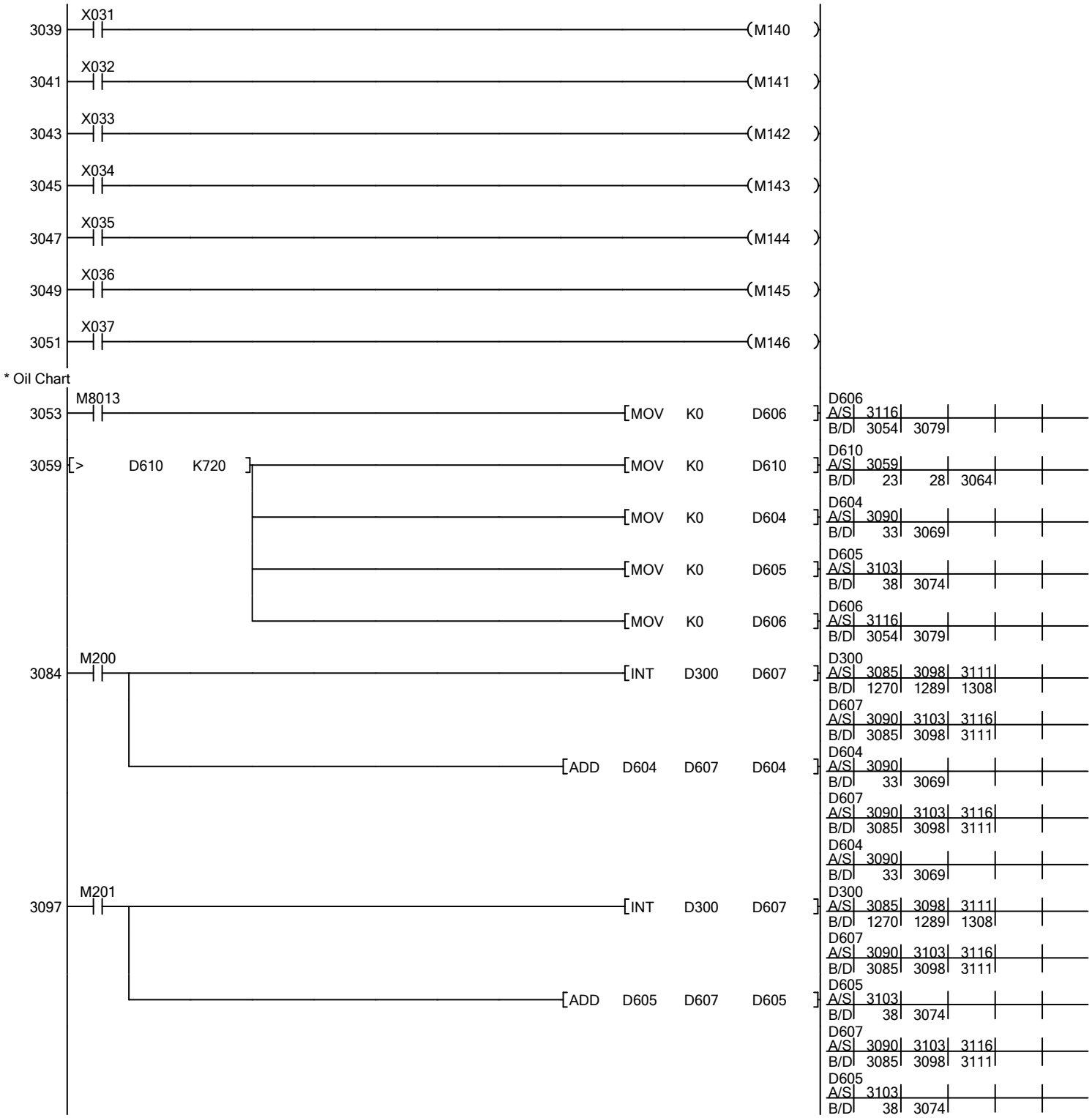

2647
M204
ADPRWK3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M201
RST M204
T16
INC D304
RST T16
2680
M205
ADPRWK3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M202
RST M205
T16
INC D304
RST T16
D550 A/S 2615 2648 2681 B/D 72
T16 A/S 2641 2674 2707 B/D 2645 2678 2711
M206 A/S 2631 2664 2697 2713 2793 A/S
M201 A/S 2493 2538 2798 2822 3097 B/D 2485 2498 2566 2671 2829
M201 A/S B/D 2856
M204 A/S 2559 2647 B/D 2672
D304 A/S 1472 2633 2666 2699 B/D 77 2642 2675 2708
T16 A/S 2641 2674 2707 B/D 2645 2678 2711
D550 A/S 2615 2648 2681 B/D 72
T16 A/S 2641 2674 2707 B/D 2645 2678 2711
M206 A/S 2631 2664 2697 2713 2793 A/S
M202 A/S 2499 2576 2802 2836 3110 B/D 2486 2492 2604 2704 2843
M202 A/S B/D 2857
M205 A/S 2597 2680 B/D 2705
D304 A/S 1472 2633 2666 2699 B/D 77 2642 2675 2708
T16 A/S 2641 2674 2707 B/D 2645 2678 2711

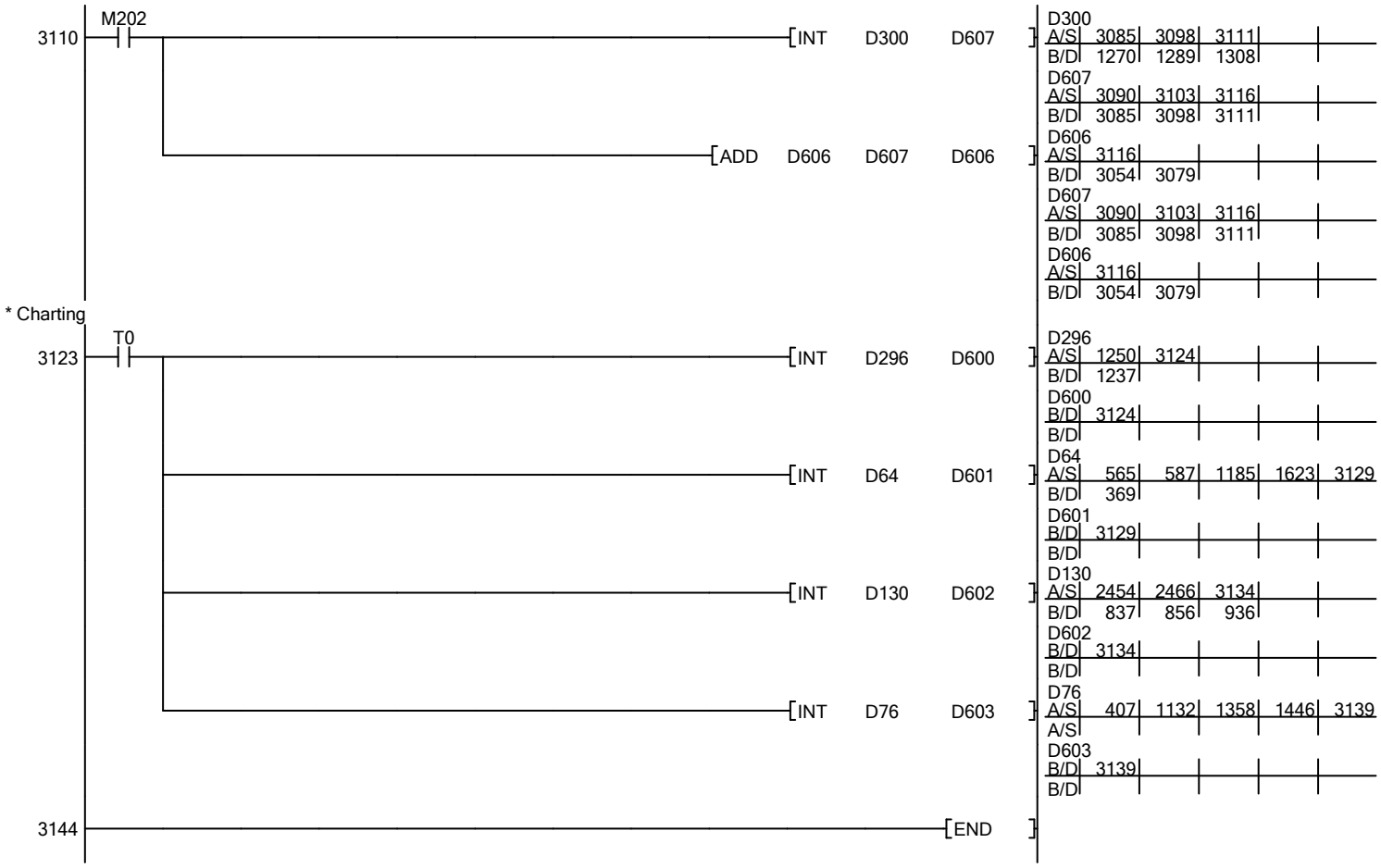

* Oil Distribuition Calculation
2713
M206
FLT D360 D360
DEDIV D42 D360 D306
DESUB E40 D220 D308
DEMUL D500 D308 D310
DEXP D310 D312
DEMUL D306 D312 D314
DEMUL D314 D502 D316
D360 A/S 2505 2543 2581 2714 2719
D42 A/S 2719 B/D 324
D306 A/S 1384 2767 B/D 2719
D220 A/S 617 626 951 1021 1030 B/D 587
D220 A/S 2732 B/D
D308 A/S 1345 1420 2745 B/D 1332 1407 2732
D500 A/S 2745
D310 A/S 1371 1433 2758 B/D 1345 1420 2745
D312 A/S 1371 1459 2767 B/D 1358 1433 2758
D314 A/S 1384 1459 2780 B/D 1371 1446 2767
D502 A/S 2780
D316 A/S 1472 B/D 1459 2780 2811 2816 2825
D316 A/S B/D 2830 2839 2844 2850
* Oil Distribution Gating
2793
M206 M200 Y022
M201 Y021
Y024
M202 Y021
Y023


* Oil Distribution Timing
2805
M11
M65
M200
X014
Y005
D316
T100
T100
A/S 2849
A/S
X017
RST M200
M200
A/S 2487 2500 2795 2808 3084
B/D 2491 2497 2528 2638 2815
M200
A/S
B/D 2855
MOV K0 D316
D316
A/S 1472
B/D 1459 2780 2811 2816 2825
D316
A/S
B/D 2830 2839 2844 2850
M201
X015
Y005
D316
T100
T100
A/S 2849
A/S
X020
RST M201
M201
A/S 2493 2538 2798 2822 3097
B/D 2485 2498 2566 2671 2829
M201
A/S
B/D 2856
MOV K0 D316
D316
A/S 1472
B/D 1459 2780 2811 2816 2825
D316
A/S
B/D 2830 2839 2844 2850
M202
X016
Y005
D316
T100
T100
A/S 2849
A/S
X021
RST M202
M202
A/S 2499 2576 2802 2836 3110
B/D 2486 2492 2604 2704 2843
M202
A/S
B/D 2857
MOV K0 D316
D316
A/S 1472
B/D 1459 2780 2811 2816 2825
D316
A/S
B/D 2830 2839 2844 2850
2849
T100
MOV K0 D316
D316
A/S 1472
B/D 1459 2780 2811 2816 2825
D316
A/S
B/D 2830 2839 2844 2850
RST M200
M200
A/S 2487 2500 2795 2808 3084
B/D 2491 2497 2528 2638 2815
M200
A/S
B/D 2855
RST M201
M201
A/S 2493 2538 2798 2822 3097
B/D 2485 2498 2566 2671 2829
M201
A/S
B/D 2856
RST M202
M202
A/S 2499 2576 2802 2836 3110
B/D 2486 2492 2604 2704 2843
M202
A/S
B/D 2857


2858
M12 X003
Y005
* Jacket Pump
2861
M11
M12 X004
Y006
* AODD Pump
2866
M11
D> D204 K30
D> D206 K30
D> D208 K30
Y007
2896
M12 X005
Y007
* Manual Pump Mode
2899
M300 M301 M302 M303 M12
Y004
M304
A/S 1730
A/S
2929
Y003
M305
A/S 1733
2964
Y002
M306
A/S 1736
* HMI Display for Inputs
2989
X000
M115


2991 X001 M116
2993 X002 M117
2995 X003 M118
2997 X004 M119
2999 X005 M120
3001 X006 M121
3003 X007 M122
3005 X010 M123
3007 X011 M124
3009 X012 M125
3011 X013 M126
3013 X014 M127
3015 X015 M128
3017 X016 M129
3019 X017 M130
3021 X020 M131
3023 X021 M132
3025 X022 M133
3027 X023 M134
3029 X024 M135
3031 X025 M136
3033 X026 M137
3035 X027 M138
3037 X030 M139
M117
A/S 1867
A/S


3039 X031 (M140)
3041 X032 (M141)
3043 X033 (M142)
3045 X034 (M143)
3047 X035 (M144)
3049 X036 (M145)
3051 X037 (M146)
* Oil Chart
3053 M8013 [MOV K0 D606]
3059 [> D610 K720] [MOV K0 D610] [MOV K0 D604] [MOV K0 D605] [MOV K0 D606]
3084 M200 [INT D300 D607] [ADD D604 D607 D604]
3097 M201 [INT D300 D607] [ADD D605 D607 D605]
D606 A/S 3116 B/D 3054 3079
D610 A/S 3059 B/D 23 28 3064
D604 A/S 3090 B/D 33 3069
D605 A/S 3103 B/D 38 3074
D606 A/S 3116 B/D 3054 3079
D300 A/S 3085 3098 3111 B/D 1270 1289 1308
D607 A/S 3090 3103 3116 B/D 3085 3098 3111
D604 A/S 3090 B/D 33 3069
D607 A/S 3090 3103 3116 B/D 3085 3098 3111
D604 A/S 3090 B/D 33 3069
D300 A/S 3085 3098 3111 B/D 1270 1289 1308
D607 A/S 3090 3103 3116 B/D 3085 3098 3111
D605 A/S 3103 B/D 38 3074
D607 A/S 3090 3103 3116 B/D 3085 3098 3111
D605 A/S 3103 B/D 38 3074


3110
M202
INT D300 D607
ADD D606 D607 D606
* Charting
3123
T0
INT D296 D600
INT D64 D601
INT D130 D602
INT D76 D603
3144
END
D300
A/S 3085 3098 3111
B/D 1270 1289 1308
D607
A/S 3090 3103 3116
B/D 3085 3098 3111
D606
A/S 3116
B/D 3054 3079
D607
A/S 3090 3103 3116
B/D 3085 3098 3111
D606
A/S 3116
B/D 3054 3079
D296
A/S 1250 3124
B/D 1237
D600
B/D 3124
B/D
D64
A/S 565 587 1185 1623 3129
B/D 369
D601
B/D 3129
B/D
D130
A/S 2454 2466 3134
B/D 837 856 936
D602
B/D 3134
B/D
D76
A/S 407 1132 1358 1446 3139
A/S
D603
B/D 3139
B/D

Position	Device	Setting Value
(1)	T0	K10
(4)	T1	K30
(7)	T2	K10
(10)	T3	K50
(13)	T4	K110
(1488)	T5	D302
(2501)	T10	K1
(2516)	T11	K2
(2539)	T12	K1
(2554)	T13	K2
(2577)	T14	K1
(2592)	T15	K2
(2626)	T16	K2
(2659)	T16	K2
(2692)	T16	K2
(2811)	T100	D316
(2825)	T100	D316
(2839)	T100	D316

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
M0	*	*(1)		
M1	*	*(1)		
M2	*	*(1)		
M3	*	*(1)		
M4	*	*(1)		
M5	*	*(1)		
M6	*	*(1)		
M7	*	*(1)		
M8	*	*(1)		
M9	*	*(1)		
M10	*	*(1)		
M11	*	*(1)		
M12	*	*(1)		
M13	*	*(1)		
M17	*	*(1)		
M18	*	*(1)		
M19	*	*(1)		
M20	*	*(1)		
M21	*	*(1)		
M22	*	*(1)		
M23	*	*(1)		
M24	*	*(1)		
M25	*	*(1)		
M50	*	*(1)		
M100	*	*(4)		
M101	*	*(4)		
M102	*	*(4)		
M103	*	*(4)		
M104	*	*(4)		
M105	*	*(4)		
M106	*	*(4)		
M107	*	*(4)		
M108	*	*(4)		
M109	*	*(4)		
M117	*	*(2)		
M200	*	*(5)		
M201	*	*(5)		
M202	*	*(5)		
M203	*	*(2)		
M204	*	*(2)		
M205	*	*(2)		
M206	*	*(3)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
M304	*	*(1)		
M305	*	*(1)		
M306	*	*(1)		
D0	*	*(2)		
D1	*	*(1)		
D2	*	*(2)		
D12	*	*(1)		
D13	*	*(1)		
D20	*	*(2)		
D21	*	*(1)		
D22	*	*(1)		
D23	*	*(1)		
D42	*	*(1)		
D43	*	*(1)		
D44	*	*(2)		
D45	*	*(2)		
D64	*	*(1)		
D65	*	*(1)		
D100	*	*(1)		
D101	*	*(1)		
D102	*	*(1)		
D103	*	*(1)		
D104	*	*(1)		
D105	*	*(1)		
D106	*	*(1)		
D107	*	*(1)		
D108	*	*(1)		
D109	*	*(1)		
D110	*	*(1)		
D111	*	*(1)		
D112	*	*(1)		
D113	*	*(1)		
D114	*	*(2)		
D115	*	*(2)		
D116	*	*(2)		
D117	*	*(2)		
D118	*	*(2)		
D119	*	*(2)		
D120	*	*(2)		
D121	*	*(2)		
D122	*	*(1)		
D123	*	*(1)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
D130	*	*(3)		
D131	*	*(3)		
D132	*	*(1)		
D133	*	*(1)		
D134	*	*(1)		
D135	*	*(1)		
D136	*	*(1)		
D137	*	*(1)		
D138	*	*(1)		
D139	*	*(1)		
D200	*	*(1)		
D220	*	*(1)		
D221	*	*(1)		
D280	*	*(1)		
D281	*	*(1)		
D282	*	*(1)		
D283	*	*(1)		
D284	*	*(1)		
D285	*	*(1)		
D286	*	*(1)		
D287	*	*(1)		
D288	*	*(1)		
D289	*	*(1)		
D290	*	*(1)		
D291	*	*(1)		
D292	*	*(1)		
D293	*	*(1)		
D294	*	*(1)		
D295	*	*(1)		
D296	*	*(1)		
D297	*	*(1)		
D298	*	*(3)		
D299	*	*(3)		
D300	*	*(3)		
D301	*	*(3)		
D302	*	*(3)		
D303	*	*(7)		
D304	*	*(4)		
D306	*	*(1)		
D307	*	*(1)		
D308	*	*(3)		
D309	*	*(3)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
D310	*	*(3)		
D311	*	*(3)		
D312	*	*(3)		
D313	*	*(3)		
D314	*	*(3)		
D315	*	*(3)		
D316	*	*(6)		
D317	*	*(2)		
D318	*	*(1)		
D319	*	*(1)		
D320	*	*(1)		
D321	*	*(1)		
D322	*	*(1)		
D323	*	*(1)		
D324	*	*(1)		
D325	*	*(1)		
D326	*	*(1)		
D327	*	*(1)		
D328	*	*(1)		
D329	*	*(1)		
D330	*	*(1)		
D331	*	*(1)		
D332	*	*(1)		
D333	*	*(1)		
D334	*	*(1)		
D335	*	*(1)		
D350	*	*(8)		
D360	*	*(1)		
D361	*	*(1)		
D550	*	*(1)		
D604	*	*(3)		
D605	*	*(3)		
D606	*	*(3)		
D607	*	*(3)		
D610	*	*(3)		
T0	*	*(1)		
T1	*	*(1)		
T2	*	*(2)		
T3	*	*(2)		
T4	*	*(1)		
T5	*	*(1)		
T10	*	*(2)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
T11	*	*(2)		
T12	*	*(2)		
T14	*	*(2)		
T16	*	*(6)		
T100	*	*(3)		
Z0	*	*(3)		
Z1	*	*(3)		
Z2	*	*(3)		
Z3	*	*(4)		
Z4	*	*(3)		
Z5	*	*(3)		
P0	*	*(1)		
P1	*	*(1)		
P2	*	*(1)		
P3	*	*(1)		
P4	*	*(1)		
P5	*	*(1)		

[Line Statement List]

Step	Line Statement
0	Timers
22	Load Once
43	E-Stops
57	Automation
66	Declarations
134	Analog to Digital
168	ADC Float
310	Flowmeter to RPM
406	Level % Converter
615	Error Threshold
826	Water Temperature Target Logic
949	Heater latching
1121	Manual Heater Control
1131	Height to Mass
1184	Dynamic Upper Thermal Boundary
1260	PO Designation
1321	PO Timer (D302)
1485	PO Dispensing
1503	Tank Scoring
1621	Need Scheduler
1656	Scheduler Cases
1674	Mode Scheduler
1728	Manual Mode Scheduler
1738	Mode Indiacator
1762	LSM Scheduler
1928	HSM Scheduler
2234	FSM Scheduler
2408	Scheduler Gates
2452	Solar Gate
2477	Manual Solar Gate
2481	Oil Scheduler
2713	Oil Distribuition Calculation
2793	Oil Distribution Gating
2805	Oil Distribution Timing
2861	Jacket Pump
2866	AODD Pump
2899	Manual Pump Mode
2989	HMI Display for Inputs
3053	Oil Chart
3123	Charting

[P Statement List]

Device	P Statement
P0	RTD to Temperature
P1	Ultrasonic to Level
P2	Sensor Averaging
P4	Sensor Logging (5 sec)

Data Name : Project Contents List

Workspace Name :
Project Name : Main_2
Title :

Data Name	Last Change	Title
Parameter	9/6/2026 12:12:22 PM	
PLC Parameter	9/6/2026 12:12:22 PM	
Network Parameter	9/6/2026 12:12:22 PM	
CC-Link	9/6/2026 12:12:22 PM	
Global Device Comment	9/6/2026 12:12:23 PM	
Program Setting		
Execution Program		
MAIN	9/6/2026 12:12:23 PM	
MAIN	9/9/2026 1:46:22 PM	
POU		
Program	9/6/2026 12:12:23 PM	
MAIN	9/9/2026 1:46:22 PM	
Local Device Comment		
Device Memory	9/6/2026 12:12:37 PM	
MAIN	9/6/2026 12:12:37 PM	